Project AegisTrader

Ideation Session Capsule

"Soul powered by backtest." — AegisTrader is a platform for growing, experimenting on, and coaching trading personalities.

Session: ideation-aegistrader-20260308-102717

Date: 2026-03-08

LAYER 1

Overview

Table of contents and content inventory

Table of Contents

Content Inventory

Category	Artifact	Description
Vision	VISION_aegistrader.md	Consolidated vision document — source of truth for production
Brief	BRIEF_soul-as-product.md	Soul as the Product positioning and onboarding
Brief	BRIEF_counterfactual-forking.md	Counterfactual Soul Forking with Trauma Test MVP
Brief	BRIEF_coaching-ux.md	User-as-Coach with Soul Studio
Research	COMPETITIVE_LANDSCAPE.md	Initial competitive analysis
Research	RESEARCH_competitive-landscape.md	Deep competitive landscape with sources
Idea Report	IDEA_soul-as-product.md	Full idea report: Soul as the Product
Idea Report	IDEA_counterfactual-forking.md	Full idea report: Counterfactual Forking (early)
Idea Report	IDEA_counterfactual-soul-forking.md	Full idea report: Counterfactual Soul Forking (detailed)
Idea Report	IDEA_coaching-ux.md	Full idea report: Coaching UX
Idea Report	IDEA_user-as-coach-soul-studio.md	Full idea report: User-as-Coach with Soul Studio
Process	ideation-graph.md	Thread registry, connections, tensions, convergence signals
Process	SNAPSHOT_01-05.md	5 session snapshots capturing dialogue state
Process	SESSION_SUMMARY.md	Final session summary
Source	request.md	Original user request
Source	manifest.md	Source materials manifest

LAYER 2

Vision

The consolidated vision document — source of truth for production

Vision: AegisTrader

What This Is

Consolidated output of ideation session `ideation-aegistrader-20260308-102717` (2026-03-08). This is the **source of truth for the production phase**. The architecture defined in Concept_1.md and PRD.md remains the technical foundation — this document reshapes the product framing, UX, and user experience layer on top of it.

Core Thesis

AegisTrader is not a backtesting platform with AI features. It is a platform for growing, experimenting on, and coaching trading personalities.

The backtest engine provides credibility. The soul provides the experience. Users choose a trading philosophy, give it life experience through historical markets, watch it develop beliefs and doctrine, fork it into alternate timelines, and coach it toward better judgment.

This is a new product category. No existing platform — not QuantConnect, not Composer, not Numerai, not any AI agent framework — offers evolving trading personalities grounded in deterministic, reproducible simulation.

Governing Principle

"Soul powered by backtest."

Every soul claim must trace back to deterministic, reproducible evidence. The deterministic engine provides reproducibility, auditability, and trust. The soul provides meaning, engagement, and differentiation. Neither works without the other.

Three Moves

Move 1: Soul as the Product

Reframe AegisTrader's positioning. The trading soul — not the backtest engine — is the primary object of user attention. "Grow trading personalities that learn from experience" is the pitch, not "backtesting platform with AI agents."

Soul-First Onboarding:

1. **Archetype picker** (min 0-2): User chooses a trading philosophy — Mean Reversion, Trend Following, Event-Driven, Defensive Value. Each archetype = pre-built strategy.md + seed soul.md. The user is choosing a worldview, not configuring software. Archetypes use trading philosophy language, not character class language.
1. **Formative experience** (min 2-5): System auto-selects a dramatic historical period and runs a compressed 3-6 month backtest. UI emphasizes soul formation: "Your agent just experienced its first 30% drawdown. It's forming its first scar tissue."
1. **Soul diff** (min 5-10): Before/after screen showing how the agent changed. Every belief is evidence-linked: "This agent now reduces size after consecutive losses — that came from losing 12% in week 3."
1. **Fork prompt**: "Want to keep training? Or fork this soul and see what happens if it had a different experience?" Bridges directly to counterfactual forking.

"Grow trading personalities" positioning: The current PRD reads like "QuantConnect plus ChatGPT." The soul-first framing creates a product category that does not exist. Competitive research confirms zero market analogs for evolving agent doctrine as a product feature.

Critical constraint (Grounder): Seed archetype quality is non-negotiable. Each archetype's strategy.md must be a genuinely viable trading approach, not a demo. A bad first backtest kills the onboarding. Evidence-linking in soul diffs is what prevents the soul-first framing from being perceived as a gimmick.

Move 2: Counterfactual Soul Forking

Fork an agent's beliefs — not just its trades — to run "what if" experiments on trading psychology itself. The branch DAG becomes a psychology lab. The soul becomes a research subject the user experiments on.

MVP Feature — Trauma Test:

- Clone the same archetype. Run one through a bull market era, one through a bear market era. Compare the resulting souls side by side.
- The bull-market agent becomes aggressive and naive about drawdowns. The bear-forged agent becomes cautious and misses rallies. Neither is "right."

- The divergence teaches how formative experience shapes trading doctrine.
- Implementation: same strategy, different time ranges, compare resulting souls. Builds on existing branch DAG.

Signature UX — Side-by-Side Soul Comparison:

- Not two separate soul pages. A unified view: "Bull-market you believes X. Crash-market you believes Y. Here's where they diverge."
- Top 3-5 most significant divergences surfaced prominently, not a raw dump. Evidence links on every belief.
- This screen is where AegisTrader's identity crystallizes for the user (see "Convergence Screen" below).

V2 Feature — Selective Amnesia:

- "What if my agent forgot its worst month?" Remove a period from a mature soul's experience. Re-derive the soul from branch history minus the excluded period.
- Tests whether specific scars are wisdom or damage. "Which scars are useful and which are just damage?" Deep emotional resonance.
- Implementation: non-trivial. Beliefs are interconnected. Requires re-derivation from parameterized experience set.

Critical constraint (Grounder): LLM non-determinism could undermine counterfactual claims. The same agent run through the same period twice may produce different soul reflections due to LLM variance, not genuine experience-driven divergence. Mitigation: may need to run each condition multiple times and surface only stable, reproducible differences.

Architecture impact:

- New entity: "soul fork" (distinct from branch — branch = alternate trades within a run, soul fork = alternate life experiences across runs)
- Soul comparison infrastructure with divergence ranking/prioritization
- Experience-set parameterization for soul derivation
- Multi-run stability checking for counterfactual claims

Move 3: User as Coach

Replace the PRD's binary approve/reject model for soul modifications with a coaching paradigm. The user is a coach developing a trading personality, not an admin approving file changes.

Three Coaching Modes:

1. **"Let it Learn" (Passive) — MVP**: Post-game film review. User reviews soul diffs after runs, accepts or flags specific beliefs. Reframe: "reviewing a lesson plan," not "approving a patch."

1. **"Guided Reflection" (Active) — v1.5**: User provides natural language coaching before soul update commits. "You're over-weighting the losses in week 3." Agent re-generates incorporating coaching. This is where most users find the most value. Recommended for v1.5, not v2.

1. **"Soul Surgery" (Direct) — MVP**: Power-user direct editing of soul.md/soul.json. Edits source-tagged as `coach_override`.

Soul Studio UI (full version v2, simplified MVP):

- **Left: Soul Timeline** — Version history color-coded by source (self-learned = blue, coached = green, counterfactual = purple). Aggressive filtering needed.
- **Center: Belief Cards** — Current soul as interactive cards with confidence, evidence count, and "Challenge" button.
- **Right: Evidence Drawer** — Supporting trades, branches, runs for any selected belief.

Belief Provenance Tagging (from day one): Every belief carries a `source` field: `self_learned`, `coach_override`, `coach_guided`, `counterfactual_derived`. This lets the agent distinguish coached beliefs from organic ones and track provenance over the soul's lifetime.

Retention Hook — "Am I a Good Coach?": Track coached beliefs vs organic beliefs over time. Surface coaching effectiveness. Per-belief attribution shows which user intuitions are correct. Turns engagement from passive observation into a trackable skill.

Critical constraints (Grounder):

- Full Soul Studio three-panel layout is too heavy for MVP. Decompose progressively — start with simplified belief cards and soul diff review, add the full layout in v2.
- Source tagging in soul.json is the non-negotiable day-one decision. Without `source` fields from the start, retrofitting provenance is painful.
- The agent must push back on coaching. If the agent rubber-stamps every user override, users lose trust. The agent should respond: "Evidence from runs #23-#31 supports my current belief. Are you sure you want to override?" Coaching is a dialogue, not a command.

How They Fit Together

Choose a personality -> Grow it through experimentation -> Coach it to improve.

```
Archetype Picker (Move 1)
     |
     v
Formative Experience Backtest (Move 1)
     |
     v
Soul Diff Screen (Move 1) <--- Evidence-grounded beliefs
     |
     +---> Fork Prompt (Move 2) ---> Trauma Test ---> Side-by-Side Comparison
     |
     v
Coaching Interaction (Move 3)
     |
     +---> Let it Learn (review)
     +---> Guided Reflection (shape)
     +---> Soul Surgery (edit)
     |
     v
Next Run / Next Fork / Next Coaching Cycle
```

The Convergence Screen: Side-by-Side Soul Comparison

The side-by-side soul comparison screen is where all three moves converge into a single experience. It is the screen where AegisTrader's identity crystallizes for the user.

On this screen, the user simultaneously:

- **Sees two souls shaped by different experiences** (counterfactual forking) — the bull-market agent and the bear-forged agent side by side
- **Reads their divergent beliefs with evidence links** (soul-as-product) — each belief traces to specific trades and runs, grounded in the deterministic engine
- **Can challenge or coach either one** (coaching UX) — the user intervenes in the soul's development, shaping which lessons stick

No other screen in the product brings all three moves together. This is the moment where "grow trading personalities through experimentation and coaching" stops being a description and becomes something the user is doing.

Design priority: This screen should receive outsized design investment. It is not a comparison utility — it is the product's defining interaction.

Key Design Decisions

1. Evidence Chain as Shared Infrastructure

All three moves depend on a common capability: **structured evidence provenance for every belief in soul.json**. This is the architectural foundation that makes soul diffs trustworthy, counterfactual comparisons meaningful, and coaching attribution possible.

Move	What it needs from evidence chain
Soul Diffs (Move 1)	Verifiable links from each belief to the trades/runs that produced it
Counterfactual Comparison (Move 2)	Distinguishing genuine experience-driven divergence from LLM noise
Coaching Attribution (Move 3)	Measuring whether coached beliefs outperform self-learned ones

Required `evidence_refs[]` structure in soul.json:

The current Concept_1.md defines `evidence_refs[]` as a flat list. This is insufficient. Each evidence reference needs:

- `run_id` / `branch_id` — which run or branch produced this evidence
- `trade_ids[]` — which specific trades within that run support the belief
- `branch_comparisons[]` — which branch-vs-branch comparisons tested the belief
- `confidence` — confidence level based on sample size (how many runs/trades support this)
- `source_type` — `self_learned` | `coach_override` | `coach_guided` | `counterfactual_derived`
- `timestamp` — when this evidence was incorporated

MVP requirement: The evidence chain schema must ship in Phase 1. Retrofitting structured provenance onto flat evidence lists is architecturally painful. This is a day-one decision.

2. Deterministic/Non-Deterministic Boundary Stays Sharp

The two-layer architecture from Concept_1.md is preserved. The soul-first framing makes the deterministic engine *more* important, not less — it is the credibility layer. The non-deterministic layer (LLM reflections, soul generation, coaching integration) may only produce proposals, annotations, hypotheses, and summaries. The deterministic layer alone produces accepted decisions, account state, fills, and branch outcomes.

The coaching paradigm (Move 3) does not blur this boundary. Coaching operates entirely within the non-deterministic layer. When a coached belief needs to become a hard deterministic constraint, it flows through the existing approval-gated write mechanism to strategy.py.

3. Source Tagging From Day One

Every belief in soul.json carries a `source` field from MVP launch. This is non-negotiable. It enables: agent pushback on coaching, coaching effectiveness measurement, counterfactual provenance, and soul integrity over time.

Boundaries

- **Not a brokerage.** AegisTrader does not execute real-money trades in initial versions.
- **Not financial advice.** The platform must clearly label itself as paper trading and research tooling.
- **Not an opaque AI black box.** Every soul belief links to evidence. Every decision is inspectable. The soul is interpretation, not truth.
- **The soul is not the source of truth.** The deterministic engine is. The soul is a learned interpretation of truth — derived, versioned, diffable, and replay-linked.

Open Questions

1. **LLM non-determinism in counterfactuals**: How many runs are needed to establish stable soul divergences? What's the threshold for surfacing a difference as "real" vs noise? This needs empirical testing during Phase 1.

1. **Guided Reflection quality**: Can LLMs reliably integrate vague natural language coaching ("you're too cautious") into coherent, structured soul updates? This determines whether v1.5 ships as designed or needs significant prompt engineering investment.

1. **Soul Timeline noise management**: After 20+ runs, the soul timeline will be dense. What summarization and grouping strategies keep it usable? Must ship with filtering from day one, not retrofit later.

1. **Archetype design**: Who designs the seed strategy.md + soul.md for each archetype? These must be genuinely viable strategies — they are the product's first impression.

1. **Coaching effectiveness validity**: How much run history is needed before coaching effectiveness metrics are statistically meaningful? Early results may be noisy.

Impact on Existing PRD

Rewrite Required

PRD Section	Change
Executive Summary	Lead with soul evolution, not backtesting
Vision	"The best platform for growing AI trading personalities"
User Stories	Soul-curation as primary stories; backtesting as supporting
Onboarding (Workflow A)	Replace create-agent-write-strategy with archetype picker flow
Section 4.2 (Approval-Gated Writes)	Expand to coaching modes for soul artifacts
Section 6 (UI)	Soul diff as hero screen; add Soul Studio; add side-by-side soul comparison
Success Metrics	Add soul-curation and coaching effectiveness metrics

Preserve As-Is

PRD Section	Reason
Deterministic engine architecture	This IS the credibility layer — soul-first framing makes it more important, not less
Branch DAG design	Reframed as biography, but same technical model
Data sources and model stack	Unchanged
File classification policy	Unchanged
Non-functional requirements	Unchanged
Live paper trading	Unchanged
Scheduling	Unchanged

New Elements to Add

Element	Scope
Archetype templates (strategy.md + seed soul.md per archetype)	MVP
Soul fork entity in data model	MVP
Structured `evidence_refs[]` in soul.json	MVP
`source` field in soul.json schema	MVP
Side-by-side soul comparison screen	MVP
CoachingInput, BeliefProvenance data entities	MVP
Guided Reflection mode	v1.5
Full Soul Studio three-panel UI	v2
Selective Amnesia counterfactual	v2
Coaching effectiveness metrics and dashboard	v2

Revised Phasing

The existing PRD phases remain valid. The soul-first framing reshapes what ships in each phase and adds a v1.5 milestone.

Phase 1: Single-Agent Soul Evolution (MVP)

Everything in the current Phase 1, plus:

- Archetype picker with 4 trading philosophies
- Compressed formative experience backtest
- Soul diff as hero screen with evidence links
- Trauma Test counterfactual (same agent, different eras)
- Side-by-side soul comparison with divergence ranking
- Mode 1 (passive coaching) + Mode 3 (soul surgery)
- Source tagging and structured evidence_refs in soul.json

Phase 1.5: Guided Coaching (NEW)

- Mode 2 (guided reflection) — natural language coaching
- Challenge button on belief cards
- Enhanced Soul Studio layout

Phase 2: Multi-Agent Competition

As currently defined in PRD, plus:

- Coaching effectiveness tracking across agents
- Per-agent soul comparison

Phase 3: Live Paper Trading

As currently defined in PRD, plus:

- Live soul updates with coaching review before commit
- Coaching effectiveness dashboard

Phase 4: Governance and Advanced Features

As currently defined, plus:

- Selective Amnesia counterfactual
- Full coaching effectiveness analytics
- Consider: Adversarial Soul Dynamics (parked from ideation — evaluate based on user demand)

Deferred Ideas

Idea	Reason Deferred	Revisit When
Adversarial Soul Dynamics	Premature complexity; multi-agent basics need to work first	Phase 3/4, if users request
Cross-Training Counterfactual	Unclear user value; academic curiosity	When users request cross-asset experiments
Broad Belief Graduation (soul auto-feeding deterministic layer)	Risks eroding the deterministic boundary	If coaching UX proves the narrow version works safely

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Competitive Positioning

AegisTrader's moat is the combination of four things no competitor offers together:

1. **Deterministic simulation truth** — reproducible, auditable backtests (validated by NautilusTrader pattern)

2. **Evolving trading souls** — versioned, diffable, evidence-linked agent doctrine (zero market analogs)
3. **Branch DAG as biography** — git-like history for both trades and beliefs (zero market analogs)
4. **Coaching relationship** — user shapes agent development through review, guidance, and direct intervention

The positioning should lead with what is unique: "Grow trading personalities that learn from experience." The backtesting engine, while essential, is table stakes — it is the credibility layer, not the differentiator.

Market context: Algorithmic trading market $24B (2025) projected to $44.5B (2030), 13.2% CAGR. AegisTrader targets the research-oriented builder segment with a product that no existing platform addresses.

Key Design Principles

1. **"Soul powered by backtest"** — the engine is credibility, the soul is experience. Neither without the other.
2. **Evidence-grounded, not narrative-first** — every soul belief links to specific trades, branches, and outcomes. Compelling AND trustworthy.
3. **Trading philosophy language, not game language** — archetypes are Mean Reversion, not Patient Contrarian. The product is serious but accessible.
4. **Coach, not approver** — the user's relationship to the soul is developmental, not bureaucratic.
5. **The soul diff is the hero screen** — after every run, the most important thing is who the agent became, not what its PnL was.
6. **Provenance is a first-class concept** — every belief knows where it came from. This enables trust, debugging, and coaching effectiveness.
7. **The agent pushes back** — coaching is a dialogue, not a command. Rubber-stamping kills trust.
8. **Progressive disclosure, not feature overload** — start simple, layer complexity across versions.

Session Provenance

This vision document was produced from ideation session `ideation-aegistrader-20260308-102717`.

Participants: Free Thinker (divergent ideation), Grounder (convergent refinement), Explorer (competitive research), Writer (observation and synthesis).

Source documents: `docs/Concept_1.md`, `docs/PRD.md`

Session artifacts: Ideation graph, 5 snapshots, 5 idea reports, 3 briefs, 2 research reports, session summary.

All session artifacts: `ideations/ideation-aegistrader-20260308-102717/session/`

LAYER 3

Exploration

Idea briefs and competitive research

BRIEF: BRIEF_COACHING-UX

Idea Brief: User-as-Coach with Soul Studio

Elevator Pitch

Replace the PRD's binary approve/reject model with a coaching paradigm. The user is a coach developing a trading personality, not an admin approving file changes. Introduce Soul Studio as the primary UI for interacting with agent beliefs.

The Gap

The PRD designs how the soul learns but not how the user shapes the soul. Binary approval works for code changes to strategy.py. For soul evolution, it is too blunt. The soul is not right or wrong — it is developing a worldview. The coaching UX must support nuanced intervention.

Three Coaching Modes

Mode 1: "Let it Learn" (Passive) — MVP

Post-game film review. Agent completes a run, generates soul update proposal. User reviews soul diff with evidence links. Accepts the full update, or flags specific beliefs for revision. Reframe: "reviewing a lesson plan," not "approving a patch."

Mode 2: "Guided Reflection" (Active) — v1.5

User provides natural language coaching before soul update commits. Examples: "You're over-weighting the losses in week 3 — that was an anomaly." Agent re-generates the soul update incorporating coaching. The user shapes the reflection, not the conclusion. This is where most users find the most value. Recommended for v1.5, not v2.

Mode 3: "Soul Surgery" (Direct) — MVP

Power-user direct editing of soul.md/soul.json. Manual edits source-tagged as "coach_override" so the agent distinguishes coached beliefs from self-learned ones.

Soul Studio UI

Three-panel layout:

- **Left: Soul Timeline** — Version history color-coded by source (self-learned = blue, coached = green, counterfactual = purple). Needs aggressive filtering from day one.
- **Center: Belief Cards** — Current soul as interactive cards. Each shows belief, confidence, evidence count, and a "Challenge" button for coaching intervention.
- **Right: Evidence Drawer** — Supporting trades, branches, runs, and PnL impact for any selected belief.

Source Tagging (soul.json)

Every belief carries a `source` field: `self_learned`, `coach_override`, `coach_guided`, `counterfactual_derived`. Architecturally important — lets the agent weight beliefs differently and maintain soul integrity as coaching accumulates.

Retention Hook: "Am I a Good Coach?"

Track coached beliefs vs organic beliefs over time. Surface coaching effectiveness: "Your coaching improved returns by 12% over what the agent would have learned on its own." Per-belief attribution shows which user intuitions are actually correct. Turns engagement from passive observation into a trackable skill.

PRD Impact

- Expand Section 4.2 (Approval-Gated Writes) to include coaching modes for soul artifacts
- Add Soul Studio as primary UI surface in Section 6
- Add `source` field to soul.json schema from day one
- Add coaching effectiveness to success metrics
- New data entities: CoachingInput, BeliefProvenance, CoachingEffectivenessMetric

MVP Scope

- Mode 1 (Passive) + Mode 3 (Soul Surgery)
- Simplified belief cards UI
- Provenance tagging (coached vs experience-derived)

v1.5 Scope

- Mode 2 (Guided Reflection) — highest value coaching mode

v2 Scope

- Full Soul Studio three-panel layout
- Challenge button on belief cards
- Coaching effectiveness dashboard

Grounder's Critical Constraints

- **Full Soul Studio is too heavy for MVP.** Decompose progressively — start with simplified belief cards and soul diff review, add the three-panel layout in v2. Don't let UI ambition delay the coaching paradigm itself.
- **Source tagging in soul.json is the non-negotiable day-one decision.** Without `source` fields from the start, retrofitting provenance is painful. This must ship in MVP regardless of which coaching modes are included.
- **The agent should push back on coaching.** If the agent rubber-stamps every user override, users lose trust in the soul's integrity. The agent must be able to respond: "Evidence from runs #23-#31 supports my current belief. Are you sure you want to override?" Coaching is a dialogue, not a command.

Risks

- Guided Reflection quality depends on LLM integrating coaching coherently
- Over-coaching may impose user biases over evidence-based learning — agent must push back (Grounder)
- Soul Surgery without guardrails could create contradictory souls
- Coaching effectiveness needs enough data history to be meaningful

Source

Grounder identified the gap ("when does the user say 'no, you learned the wrong lesson'?"). Free Thinker proposed three-mode model, Soul Studio, provenance tagging, and "am I a good coach?" retention hook. Both agents converged on coaching as the right metaphor for user-soul interaction.

Idea Brief: Counterfactual Soul Forking

Elevator Pitch

Let users fork an agent's beliefs — not just its trades — to run "what if" experiments on trading psychology itself. The branch DAG becomes a psychology lab, not just an optimization tool.

The Concept

The PRD's branch DAG forks trade decisions: "what if I used a tighter stop-loss?" Counterfactual soul forking adds a higher level: "what if this agent never experienced the 2022 crash? What kind of trader would it be?" This turns the soul from a passive accumulator into a research subject the user experiments on.

Three Experiments

1. Trauma Test (MVP)

Question: "What if my agent started in 2019 vs 2022?"

Clone the same archetype, run through different eras, compare resulting souls side by side. The bull-market agent becomes aggressive and naive about drawdowns. The bear-forged agent becomes cautious and misses rallies. Neither is "right" — the divergence teaches how experience shapes doctrine.

Implementation: Simple — same strategy, different time ranges, compare resulting souls. No new architectural concepts beyond the existing branch DAG.

Key UX: Side-by-side soul comparison screen. "Bull-market you believes X. Crash-market you believes Y. Here's where they diverge." This screen could be one of the product's most compelling views.

2. Selective Amnesia (v2)

Question: "What if my agent forgot its worst month?"

Remove a specific period from a mature soul's experience. Re-derive the soul from branch history minus the excluded period. Tests whether scars are wisdom or just damage.

Implementation: Non-trivial — beliefs are interconnected, can't just delete lines. Requires re-derivation from parameterized experience set.

Why it matters: Deep emotional resonance. Every trader has experiences that made them too cautious. Testing "what if I removed that memory?" is something no human trader can do but every trader wishes they could.

3. Cross-Training (Parked)

Expose an agent to a different asset class. Parked — too ambitious, unclear user value.

Architecture Impact

- New entity: "soul fork" (distinct from branch — branch = alternate trades, soul fork = alternate life experiences)
- Soul comparison infrastructure: diff two souls, highlight divergent beliefs with evidence chains
- Experience-set parameterization for soul derivation

Competitive Position

No trading platform, AI agent platform, or backtesting tool offers belief-level counterfactual experimentation. Zero analogs.

MVP Scope

- Trauma Test only: same archetype, different time ranges, soul comparison
- Side-by-side soul comparison screen with divergence highlighting
- "Fork this soul" prompt in onboarding flow
- Soul fork entity in data model

Grounder's Critical Constraints

- **LLM non-determinism could undermine counterfactual claims.** The same agent run through the same period twice may produce different soul reflections due to LLM variance, not genuine experience-driven divergence. Mitigation: may need to run each condition multiple times and surface only the stable, reproducible differences. This is a credibility requirement.

- **Comparison UX must surface top 3-5 divergences, not dump everything.** Raw soul diffs will be noisy. The comparison screen needs intelligent summarization — prioritize the most significant belief divergences, not an exhaustive list.

Risks

- LLM non-determinism may produce spurious divergences — need multiple runs to isolate stable differences (Grounder)
- Users may over-interpret divergences as causal when correlational
- Many soul forks could become visually overwhelming — surface top 3-5, not all (Grounder)

Source

Free Thinker proposed forking beliefs, "nature vs nurture" framing. Grounder reframed to "alternate timelines," confirmed Trauma Test as MVP, endorsed Selective Amnesia for v2, parked Cross-Training. Both converged on side-by-side comparison as signature UX.

BRIEF: BRIEF_SOUL-AS-PRODUCT

Idea Brief: Soul as the Product

Elevator Pitch

AegisTrader is not a backtesting platform with AI agents. It is a platform for growing trading personalities that learn from experience. The backtest is the gym. The soul is the athlete. The branch DAG is the biography. The deterministic engine is what makes the soul trustworthy.

The Reframe

Before (PRD)	After (Soul-First)
"Run backtests, see results"	"Grow a trading personality, watch it evolve"
Backtest results are the output	Soul evolution is the output
Branch DAG is debugging	Branch DAG is biography
PnL charts are the hero	Soul diff is the hero
User is an operator	User is a curator/coach

Key Formulation

"Soul powered by backtest." The engine provides credibility. The soul provides experience. Neither works without the other.

Soul-First Onboarding

1. **Archetype picker** (min 0-2): Choose a trading philosophy — Mean Reversion, Trend Following, Event-Driven, Defensive Value. Each = pre-built strategy.md + seed soul.md.
2. **Formative experience** (min 2-5): System runs a compressed 3-6 month backtest through a dramatic historical period. UI emphasizes soul formation, not PnL.
3. **Soul diff** (min 5-10): Before/after screen with evidence-linked beliefs. "This agent now reduces size after consecutive losses — that came from losing 12% in week 3."

4. **Fork prompt**: "Want to keep training? Or fork this soul and see what happens with a different experience?" Bridges to counterfactual forking.

Evidence-Grounded Soul Diffs

The soul diff is the hero screen but must stay evidence-grounded. Format: "Before run #47, this agent believed X. After run #47, it believes Y. Here's the trade that changed its mind." Every belief transformation links to specific trades, branches, and PnL impact.

Competitive Position

- Zero market analogs for evolving agent doctrine as a product feature
- Closest competitor (QuantConnect Mia V2) is a coding assistant, not an evolving personality
- "Grow trading personalities" creates a product category that does not exist

MVP Scope

- 4 pre-built trading philosophy archetypes
- Compressed formative experience backtest
- Soul diff as hero screen with evidence links
- Fork prompt bridging to counterfactual forking
- All deterministic engine requirements unchanged

PRD Impact

- Rewrite: Executive Summary, Vision, primary User Stories, Onboarding Workflow, UI hero screens
- Preserve: Deterministic engine architecture, Branch DAG design, Approval-gated writes, Data sources, Model stack

Grounder's Critical Constraints

- **Seed archetype quality is non-negotiable.** Each archetype's strategy.md must be a genuinely viable trading approach, not a demo. A bad first backtest undermines the entire onboarding promise. This is a hard prerequisite, not a nice-to-have.

- **Evidence-linking in soul diffs prevents "gimmick" perception.** Without concrete trade-to-belief links, the soul-first framing risks feeling like marketing rather than substance. Every belief must trace back to specific runs and trades.

Risks

- "Growing personalities" may feel gamified — mitigate with evidence grounding and domain language
- Pre-built archetypes must be genuinely useful strategies (Grounder: non-negotiable)
- Soul evolution quality depends on LLM reflection quality
- Formative experience period selection must be curated

Source

Free Thinker proposed the inversion. Grounder refined to "soul powered by backtest" and corrected archetype naming to trading philosophy language. Competitive research confirmed zero analogs. Both agents converged.

Competitive Landscape: AI Trading Research Platforms

Overview

AegisTrader sits at the intersection of algorithmic backtesting, AI agent frameworks, and trading research tools. No existing product combines all of its proposed capabilities — deterministic backtesting with AI agent exploration, branch DAG history, evolving trading souls, and multi-agent competition. Below is an analysis of the closest competitors and adjacent products.

Direct Competitors (Algorithmic Backtesting Platforms)

QuantConnect

- **What it is:** Cloud-based algorithmic trading research platform with backtesting, live trading, and a community marketplace.
- **Strengths:** Mature backtesting engine, multi-asset support (equities, options, futures, crypto, forex), LEAN open-source engine, large community, Alpha Streams marketplace.
- **Weaknesses:** Code-heavy (C#/Python), no AI agent layer, no branching/exploration model, no concept of evolving agent memory, steep learning curve for non-developers.
- **AegisTrader differentiation:** Soul evolution, branch DAG, natural language strategy definition, AI-driven exploration. QuantConnect is a quant tool; AegisTrader is an AI research lab.

Zipline (now largely community-maintained)

- **What it is:** Python backtesting framework originally from Quantopian.
- **Strengths:** Simple Python API, good for single-strategy backtests.
- **Weaknesses:** No longer actively maintained by a company, no UI, no agent layer, no live trading support, single-run linear execution only.
- **AegisTrader differentiation:** Everything — UI, agents, branching, souls, live paper trading.

Backtrader

- **What it is:** Open-source Python backtesting framework.
- **Strengths:** Flexible, supports multiple data feeds, community plugins.
- **Weaknesses:** No AI integration, no branching, no UI beyond basic plotting, developer-only tool.
- **AegisTrader differentiation:** Full platform vs. library. Agent layer, UI, and soul concepts are entirely absent here.

AlgoTrader

- **What it is:** Enterprise algorithmic trading platform for quantitative hedge funds.
- **Strengths:** Institutional-grade execution, multi-asset, FIX protocol support, risk management.
- **Weaknesses:** Enterprise pricing, no AI agent layer, designed for traditional quant workflows, not research exploration.
- **AegisTrader differentiation:** AI-native, research-first, accessible to non-institutional users.

Adjacent Competitors (AI-Assisted Trading Tools)

Composer

- **What it is:** No-code visual strategy builder for retail investors.
- **Strengths:** Beautiful UI, drag-and-drop strategy composition, automated rebalancing, live trading via brokerage integration.
- **Weaknesses:** No AI agents, no backtesting branching, no strategy evolution, rule-based only, limited to long-only equities/ETFs.
- **AegisTrader differentiation:** AI agents, branch exploration, soul evolution, depth of research capability.

TradingView

- **What it is:** Charting platform with Pine Script strategy backtesting.
- **Strengths:** Massive user base, excellent charting, social features, strategy alerts.
- **Weaknesses:** Pine Script is limited, no agent autonomy, no branching, no AI reflection, backtesting is basic.
- **AegisTrader differentiation:** Completely different category — TradingView is charting-first, AegisTrader is research-and-agent-first.

Alpaca + AI Wrappers

- **What it is:** Various projects combining Alpaca's paper trading API with LLM-based decision-making.
- **Strengths:** Simple to prototype, real paper trading fills, API-first.
- **Weaknesses:** Typically proof-of-concept quality, no deterministic replay, no branching, no soul evolution, no audit trail, no durable workflows.
- **AegisTrader differentiation:** Production-grade platform vs. weekend hack. The deterministic/non-deterministic boundary is the key missing piece in all Alpaca+AI projects.

Adjacent Competitors (AI Agent Platforms Applied to Trading)

CrewAI / AutoGen / LangGraph Agent Frameworks

- **What they are:** General-purpose multi-agent AI frameworks.
- **Strengths:** Flexible, support tool calling, some support checkpointing (LangGraph).
- **Weaknesses:** Not trading-specific, no deterministic simulation core, no financial data integration, no branch DAG model, no trading soul concept.
- **AegisTrader differentiation:** Domain-specific platform built on top of these concepts, not a general framework.

Numerai

- **What it is:** Crowdsourced hedge fund where data scientists submit predictions in tournament format.
- **Strengths:** Novel tournament/competition model, real staking mechanism, large data science community.
- **Weaknesses:** Users submit signals, not agents. No agent autonomy, no strategy evolution, no branching, black-box evaluation.
- **AegisTrader differentiation:** Personal agent evolution vs. crowdsourced signal submission. AegisTrader's multi-agent tournament is within a single user's research environment.

Emerging / Niche Players

Various "AI Trading Bot" Products

- Products like Stoic, 3Commas, Cryptohopper (mostly crypto-focused).

- Typically rule-based bots with some ML signal integration.
- No branching, no soul evolution, no deterministic audit trail.
- AegisTrader is fundamentally different in ambition and architecture.

Academic / Research Tools

- Tools like bt (Python), VectorBT, FinRL.
- Strong on backtesting performance and vectorized computation.
- No AI agent layer, no UI platform, no soul/evolution concepts.

Key Gaps in the Market That AegisTrader Fills

1. **No existing platform combines deterministic backtesting with AI agent exploration.** This is the core gap. QuantConnect has backtesting but no agents. Agent frameworks have agents but no financial simulation core.
1. **No platform offers branch DAG exploration for trading strategies.** Git-like history for trading decisions is genuinely novel. No competitor does this.
1. **No platform has "trading soul" evolution.** The concept of agents developing durable, versioned, evidence-linked doctrine over time is unique.
1. **No platform bridges natural language strategy definition with deterministic execution.** Composer does visual no-code; QuantConnect does pure code. The markdown+optional-Python model is a genuine middle ground.
1. **Multi-agent competition within a personal research environment** is not offered by any existing product. Numerai does competition across users, not within a single user's agent fleet.

Competitive Risks

1. **QuantConnect could add AI features.** They have the backtesting engine and community. If they layer on an agent framework, they could capture this space quickly.
1. **LangChain/LangGraph ecosystem could release a finance-specific template.** Given their focus on vertical applications, a "LangGraph for Trading" starter could emerge.
1. **Alpaca or other brokers could build a first-party AI research layer.** They own the execution and data; adding research UI is a smaller step for them.
1. **General AI coding agents (Devin, Cursor, etc.) could be pointed at trading codebases.** Not a direct competitor but could reduce the perceived need for a specialized platform.

Summary

AegisTrader's strongest competitive moat is the combination of: (a) deterministic simulation truth, (b) AI agent exploration and reflection, (c) branch DAG history, and (d) evolving trading souls. No single competitor offers even two of these together. The nearest threat is QuantConnect adding AI features, but their code-first culture and existing architecture would make the soul/branch/agent model a significant pivot.

The product should lean into the branch DAG and soul evolution as its primary differentiators in positioning — these are the concepts that have no existing analog in the market.

RESEARCH: RESEARCH_COMPETITIVE-LANDSCAPE

Research: Competitive Landscape of AI Trading Platforms

Requested by: team-lead **Date:** 2026-03-08

Question

What is the competitive landscape of AI trading platforms, agent-based backtesting tools, and AI-assisted strategy evolution products? Specifically: existing solutions, branch/fork concepts, agent evolution/memory, and how platforms handle the deterministic vs non-deterministic boundary.

Findings

1. QuantConnect + Mia V2

QuantConnect is the dominant open-source algo trading platform (300K+ investors, 300+ hedge funds, LEAN engine). Their 2025 release of **Mia V2** is the most relevant competitor to AegisTrader's vision:

- Mia V2 is an agentic AI assistant that can ideate strategies, write code, run backtests, and debug them through a chat interface.
- Uses MCP server integration for structured tool access to the LEAN engine.
- Wraps leading US LLM models with a cloud-only MCP to stay compliant with data licenses.
- Supports parameter sensitivity testing with heatmaps across thousands of backtest iterations.
- **Limitation vs AegisTrader:** Mia V2 is a *coding assistant* for quant strategies, not a *self-evolving agent with memory/soul*. It does not branch, fork, or maintain evolving doctrine. It helps humans write code, not explore alternate decision branches autonomously.

2. Composer

Composer is a no-code AI trading platform for retail investors:

- Natural language strategy creation, sub-second backtesting, automated execution.
- 3,000+ community-built strategies ("symphonies") with sharing/discovery.

- Supports stocks, crypto, and options with visual editing.
- $200M+ daily trading volume.
- **Limitation vs AegisTrader:** Composer is retail-focused, no-code, with no agent evolution, no branching/forking concept, no deterministic/non-deterministic separation, and no memory persistence. Strategies are static once created.

3. Alpaca (as Infrastructure)

Alpaca is the leading developer-first brokerage API:

- Historical data, real-time streams, paper trading, and live execution.
- MCP Server enabling AI agents (Claude, ChatGPT) to trade directly.
- Best broker for algorithmic trading 2026 (BrokerChooser).
- Backtrader integration available.
- **Relationship to AegisTrader:** Alpaca is infrastructure, not a competitor. AegisTrader's PRD already identifies Alpaca as the MVP data/paper-trading provider. Community projects build agents *on top of* Alpaca, but none implement the branch DAG, soul evolution, or deterministic separation that AegisTrader proposes.

4. AlgoTrader

Enterprise-grade institutional platform:

- Java-based, high-frequency capable, multi-asset.
- OMS/PMS, compliance-ready, real-time risk management.
- ML integration for strategy development and execution.
- Enterprise pricing (not retail accessible).
- **Limitation vs AegisTrader:** Purely institutional, no AI agent evolution, no branching concept, no natural-language strategy definition. Different market segment entirely.

5. NautilusTrader

The most architecturally relevant open-source project:

- Rust-native core with deterministic event-driven architecture.
- Python control plane for strategy logic and orchestration.
- Nanosecond resolution, 5M+ rows/second throughput.
- Same execution semantics in research and live modes (research-to-live parity).
- Deterministic results: identical output given same data, config, and random seed.
- Explicitly keeps AI/ML tooling out of scope to maintain core engine focus.
- **Relevance to AegisTrader:** NautilusTrader validates the architectural pattern of a deterministic engine with a separate control plane. However, it deliberately excludes AI

integration, UI dashboards, and orchestration. AegisTrader's deterministic Layer 1 should learn from NautilusTrader's approach, but the AI agent layer, soul evolution, and branch DAG are entirely additive.

6. Other Notable Platforms

Platform	Key Feature	Gap vs AegisTrader
TrendSpider	AI pattern recognition (148 candlestick patterns), point-and-click backtesting	No agent autonomy, no evolution, no branching
LuxAlgo	Natural language backtesting assistant	Shallow AI integration, no agent memory
Tickeron	34 AI stock trading systems, HFT 5min/15min agents	Pre-built models, not user-defined evolving agents
Capitalise.ai	Code-free automation from natural language	No research depth, no branching, no soul

7. Branch/Fork Concept in Trading: A White Space

No existing trading platform implements a git-like branch/fork concept for strategy versioning and exploration.

Current platforms handle strategy variants through:

- Parameter optimization grids (QuantConnect, AlgoTrader)
- Walk-forward optimization (TradeStation, MotiveWave)
- Manual strategy duplication and comparison
- Community strategy sharing (Composer)

AegisTrader's branch DAG — where agents can fork from checkpoints, explore alternate decisions, and compare outcomes with soul deltas — is genuinely novel in the trading platform space.

8. Agent Evolution/Memory: Emerging but Unproductized

The concept of persistent agent memory is a major 2025-2026 AI research theme:

- arXiv paper "Memory in the Age of AI Agents" (Dec 2025) taxonomizes factual, experiential, and working memory.
- Google Research's Titans + MIRAS treats long-term memory as a first-class design object.
- SEAL proposes models that generate their own finetuning data for persistent weight updates.
- "Digital DNA" / Constitutional Guardrails concept mirrors AegisTrader's "soul doctrine."

No trading product has productized agent memory evolution. Individual developers have built one-off agents with Alpaca that remember context, but these are demos, not platforms.

9. Deterministic vs Non-Deterministic Boundary

How existing platforms handle this:

Platform	Approach
NautilusTrader	Strict deterministic core; AI/ML explicitly out of scope
QuantConnect	Deterministic LEAN engine + Mia V2 as separate chat assistant (not in the execution path)
Composer	AI generates strategies that then execute deterministically; AI is not in the loop during execution
AlgoTrader	ML models can be *part of* the strategy (blurs the boundary)
TrendSpider	AI suggests improvements post-backtest (separated but shallow)

AegisTrader's explicit contract — where the non-deterministic layer can only produce proposals/annotations/hypotheses, and only the deterministic layer produces accepted decisions/fills/scores — is the most rigorous formulation of this boundary seen in any platform.

10. LangGraph Time-Travel: Conceptual Enabler

LangGraph's checkpoint and time-travel features are directly applicable:

- Persistent state snapshots at each execution step.
- Resume from any checkpoint with state modification.
- Every modification creates a fork in execution history.
- Audit trail of all changes.
- Production checkpointers available (PostgresSaver, SqliteSaver).

No one has applied LangGraph time-travel specifically to trading agent workflows. This is an open opportunity for AegisTrader.

Key Takeaways

1. **The branch DAG concept is a genuine white space.** No trading platform offers git-like forking of strategy execution with checkpoint-based exploration. This is AegisTrader's strongest differentiator.
1. **Agent soul/memory evolution is unproductized in trading.** The AI research community is actively working on agent memory (Dec 2025 papers), but no trading product has turned this into a feature. AegisTrader would be first-to-market.
1. **The deterministic/non-deterministic separation is AegisTrader's structural advantage.** Most platforms either exclude AI from execution entirely (NautilusTrader) or blur the boundary (AlgoTrader ML models in strategies). AegisTrader's explicit contract is more rigorous than any competitor.
1. **QuantConnect's Mia V2 is the closest competitor in spirit but fundamentally different in approach.** Mia is a coding assistant; AegisTrader agents are autonomous researchers with evolving memory. Different interaction model entirely.
1. **NautilusTrader validates the deterministic engine architecture.** Its Rust-native, event-driven, deterministic design with Python control plane is exactly the pattern AegisTrader's Layer 1 should follow. AegisTrader adds everything NautilusTrader deliberately excludes (AI agents, UI, orchestration, branching).

Sources

#	Source	URL	What It Contributed
1	QuantConnect	https://www.quantconnect.com/	Platform overview, Mia V2 capabilities
2	QuantConnect Mia V2 Announcement	https://www.quantconnect.com/announcements/19846/your-ai-quant-developer/	AI assistant architecture details
3	Composer	https://www.composer.trade/	No-code AI trading platform features
4	Composer Trade With AI	https://www.composer.trade/ai	AI strategy creation capabilities
5	Composer BusinessWire	https://www.businesswire.com/news/home/20251021050436/en/	Trade With AI tool launch details
6	Alpaca	https://alpaca.markets/	Developer API, MCP server, paper trading
7	Alpaca AI Agents Blog	https://alpaca.markets/learn/how-traders-are-using-ai-agents-to-create-trading-bots-with-alpaca	Community AI agent usage patterns
8	Alpaca Best Broker 2026	https://alpaca.markets/blog/alpaca-recognized-as-best-broker-for-algorithmic-trading-in-2026-by-brokerchooser/	Market positioning
9	AlgoTrader	https://www.algotraders.ai/	Enterprise platform capabilities
10	NautilusTrader GitHub	https://github.com/nautechsystems/nautilus_trader	Architecture, deterministic design
11	NautilusTrader Docs - Architecture	https://nautilustrader.io/docs/latest/concepts/architecture/	Detailed architecture patterns
12	NautilusTrader Docs - Backtesting	https://nautilustrader.io/docs/latest/concepts/backtesting/	Deterministic backtesting details
13	NautilusTrader Homepage	https://nautilustrader.io/	Feature overview, performance specs
14	Python Backtesting Landscape 2026	https://python.financial/	Ecosystem comparison
15	LangGraph Time Travel Concepts	https://langchain-ai.github.io/langgraph/concepts/time-travel/	Checkpoint and fork mechanics
16	LangGraph Time Travel How-To	https://docs.langchain.com/langsmith/human-in-the-loop-time-travel	Server API for time travel

#	Source	URL	What It Contributed
17	Debugging Non-Deterministic Agents (DEV)	https://dev.to/sreeni5018/debugging-non-deterministic-llm-agents-implementing-checkpoint-based-state-replay-with-langgraph-5171	Checkpoint-based state replay patterns
18	arXiv: Memory in the Age of AI Agents	https://arxiv.org/abs/2512.13564	Agent memory taxonomy and research frontiers
19	AI Predictions 2026	https://www.vastkind.com/ai-predictions-2026-memory-agents-evals/	Persistence as key agent differentiator
20	Agentic AI Maturity Model	https://dr-arsanjani.medium.com/ai-in-2026-predictions-mapped-to-the-agentic-ai-maturity-model-c6f851a40ef5	Digital DNA / Constitutional Guardrails concept
21	Algo Trading Market Report 2026	https://www.globenewswire.com/news-release/2026/03/06/3250835/0/en/	Market size ($24B 2025, $44.5B 2030)
22	TrendSpider	https://trendspider.com/product/strategy-development-and-backtesting-tools/	AI pattern recognition backtesting
23	LuxAlgo	https://www.luxalgo.com/backtesting/	Natural language backtesting
24	AI Trading Agent Medium Post	https://laurentiu-raducu.medium.com/i-created-an-ai-trading-agent-heres-what-it-did-after-one-month-3d6c54c68445	Real-world AI agent trading example

Citation Log

Searches performed:

1. "QuantConnect AI agent trading backtesting platform 2025 2026"
2. "Composer trading AI automated strategy platform features"
3. "Alpaca AI trading tools agent-based backtesting 2025 2026"
4. "AI native trading research platform agent evolution memory backtesting 2025 2026"
5. "AlgoTrader platform AI features 2025 2026 algorithmic trading"
6. "trading platform branch fork concept git-like strategy versioning backtesting"
7. "AI trading agent memory persistence evolving strategy soul doctrine 2025 2026"
8. "deterministic backtesting non-deterministic AI reasoning separation trading platform"
9. "LangGraph trading agent checkpoint time-travel fork workflow 2025 2026"
10. "NautilusTrader features architecture deterministic backtesting 2026"

LAYER 4

Origins

Original request and source materials

ORIGINAL REQUEST

User Request

Date: 2026-03-08

The user requested an ideation session using two existing documents from the `docs/` folder:

1. `Concept_1.md` — A detailed architectural concept document covering core architecture, trading soul design, branching/git-tree history, data sources, strategy definition, deterministic vs non-deterministic boundaries, technology stack, and phased build plan for a multi-agent AI trading research platform.

1. `PRD.md` — A comprehensive Product Requirements Document for "Project AegisTrader" — a multi-agent financial paper trading research platform. Covers executive summary, functional requirements, workflows, data model, UI requirements, non-functional requirements, and rollout plan.

The user wants the ideation team to explore and build on this concept.

Source Materials Manifest

Session: AegisTrader - Multi-Agent Financial Paper Trading Research Platform **Captured:** 2026-03-08

#	File	Type	Original Location
1	request.md	User request	(inline input)
2	Concept_1.md	Concept seed / Architecture document	docs/Concept_1.md
3	PRD.md	Product Requirements Document	docs/PRD.md

LAYER 5

Process

Ideation graph, snapshots, idea reports, and session summary

Ideation Graph — Project AegisTrader

Session Metadata

- **Concept**: Multi-Agent Financial Paper Trading Research Platform
- **Session ID**: ideation-aegistrader-20260308-102717
- **Date**: 2026-03-08
- **Depth**: Standard
- **Focus**: Explore new dimensions (UX paradigms, product positioning, differentiation, novel agent interaction patterns) while refining existing architectural and PRD decisions

Seed Concepts

The session begins from two source documents:

1. **Concept_1.md** — A detailed architectural blueprint covering:

- Two-layer architecture: deterministic trading engine + non-deterministic agent layer
- OpenClaw-inspired control plane pattern
- Stack: React/Vite + Node control plane + Python runtime + Temporal + Postgres + Convex + Parquet + LiteLLM
- Trading soul as structured evolving doctrine (soul.md + soul.json)
- Branch DAG for git-tree style history
- Strategy as markdown + optional Python hooks
- Deterministic vs non-deterministic contract
- Phased rollout: single-agent backtester -> multi-agent tournament -> paper live -> governance

1. **PRD.md** — Full product requirements document covering:

- 10 product principles anchored in determinism and auditability
- Agent authoring, backtesting, branching, tool calling, approval-gated writes
- Live paper trading modes (always-on, scheduled, interval, mixed)
- Competition rules with leaderboard-only visibility
- UI requirements: branch tree, trade ledger, soul viewer, approval inbox
- File classification: safe_to_edit / approval_required / never_editable_by_agent

- Data model with 17+ core entities
- 5 major workflows (create agent, backtest, branch, approval, live paper)
- MVP scope and rollout phases

Thread Registry

Thread ID	Label	Status	Origin	Forked From	Flagged
T1	Soul as the Product	**ARBITER CONFIRMED**	Free Thinker R1	seed	INTERESTING
T2	Adversarial Soul Dynamics	**parked**	Free Thinker R1	seed	--
T3	Soul Diff as First-Class UX	**active (refined)**	Free Thinker R1	seed	--
T4	Counterfactual Soul Forking	**ARBITER CONFIRMED**	Free Thinker R1	seed	INTERESTING
T5	Soul-to-Deterministic Graduation	**cautioned**	Free Thinker R1	seed	--
T6	Coaching UX (User-Soul Relationship)	**active (pending eval)**	Grounder R2	T1, T3	expected

Active Threads

T1: Soul as the Product (not the backtest)

Proposed by: Free Thinker (Round 1) **Core idea**: Invert the product framing. The backtest is the gym; the soul is the athlete. Users "grow and curate trading personalities." The branch DAG becomes the soul's biography. Reframes UX from analytics dashboard to "creature evolution simulator for finance." **Grounder response (Round 1)**: Agrees this is the strongest direction but grounds it — "soul powered by backtest, not soul instead of backtest." The deterministic engine is the credibility; the soul is the experience. Key positioning insight from Grounder: saying "backtesting platform with AI agents" makes users hear "QuantConnect-with-ChatGPT," but "you grow trading personalities that learn from experience" is a completely different product. **Round 2 development — Detailed onboarding flow (Free Thinker)**:

- **Min 0-2: Archetype picker.** User picks a trading personality (not a blank strategy.md). Each archetype = pre-built strategy.md + seed soul.md. Free Thinker proposed: Patient Contrarian, Momentum Surfer, Cautious Earner, Regime Reader.
- **Min 2-5: Formative experience.** System auto-selects a dramatic historical period (2020 crash, 2021 melt-up, 2022 bear) and runs a compressed 3-6 month backtest. UI emphasizes soul formation over PnL: "Your agent just experienced its first 30% drawdown. It's forming its first scar tissue."
- **Min 5-10: Soul diff + fork prompt.** Before/after screen: seed archetype vs shaped personality. Specific trades linked to specific beliefs. Then: "Want to keep training? Or fork this soul and see what happens if it had a different experience?" — bridging directly to T4.
- **Key insight (Free Thinker)**: "The current onboarding creates a backtesting user. This onboarding creates a soul curator."

Grounder correction (Round 2): Archetype names should be trading philosophies, not character classes. Grounder's specific alternatives: "Mean Reversion — Waits for overreactions," "Trend Follower — Rides momentum until it breaks," "Event-Driven — Trades around catalysts." Quote: "The user should feel like they're choosing a trading philosophy, not a Pokemon." **Key framing locked**: "Soul powered by backtest" — engine is credibility, soul is experience. **Status**: Converging and deepening. Onboarding flow validated. Core framing solidified.

T3: Soul Diff as First-Class UX Moment

Proposed by: Free Thinker (Round 1) **Core idea**: Elevate the soul diff from metadata to the most important screen in the product. **Grounder response (Round 1)**: Keep evidence-grounded. Grounder's specific UI direction: "show the soul diff as a before/after with evidence links. 'Before run #47, this agent believed X. After run #47, it believes Y. Here's the trade that changed its mind.'" **Status**: Active, refined. Now integrated into the onboarding flow (T1) as the payoff moment after first backtest.

T4: Counterfactual Soul Forking

Proposed by: Free Thinker (Round 1) **Core idea**: Fork the soul itself, not just trade history. Run two versions with different life experiences, observe doctrinal divergence. **Round 2 development — Three counterfactual experiments proposed (Free Thinker)**:

1. **Trauma Test** (MVP feature): Same agent, different eras. "What if your trend-follower was forged in the 2008 crash vs the 2020-2021 bull run?" The bull-baby becomes aggressive, over-leveraged, naive about drawdowns. The bear-forged agent becomes cautious, defensive, misses rallies. Neither is "right" — the divergence teaches the user how

experience shapes doctrine. Most intuitive because it maps to how humans think about trading psychology.

2. **Selective Amnesia** (compelling v2): "What if we erased the agent's worst month?" Tests whether specific experiences are load-bearing in the soul's structure. Does removing the worst month make the agent reckless, or does it remove over-indexed damage? "Which scars are useful and which are just damage?" Needs the soul to be mature enough first. **Grounder implementation note**: Can't just delete lines from soul.md — beliefs are interconnected. Would need to re-derive the soul from the branch history minus the excluded period. Architecturally non-trivial but scoped for v2.
3. **Cross-Training** (parked): "What if my equity agent lived through the crypto winter of 2022?" Expose an agent to a different asset class's history to see if trading wisdom transfers across domains. Most creative but most ambitious. Free Thinker self-assessed as too ambitious for early versions.

Grounder Round 2: Trauma Test confirmed as MVP counterfactual feature. Flagged need for **side-by-side soul comparison UX** — not two separate soul pages but a unified comparison: "Bull-market you believes X. Crash-market you believes Y. Here's where they diverge." Grounder: "That screen could be one of the most compelling things in the whole product." **Status**: Converging. Trauma Test is the concrete MVP expression. Selective Amnesia is the compelling v2 hook.

T6: Coaching UX (User-Soul Relationship) — NEW

Proposed by: Grounder (Round 2), emerging from T1/T3 discussion **Core idea**: The Grounder asked: "When does the user say 'no, you learned the wrong lesson'?" The PRD frames the user-agent relationship as approver (approve/reject file changes). But if the soul is the product, the user's relationship to the agent is more like a **coach** than a bureaucratic approver. The user guides the agent's development, corrects mistaken beliefs, reinforces good lessons. **Status**: New thread, about to be explored by Free Thinker. **Note**: This reframes Workflow D (Approval-Gated File Change) from a permission system into a coaching interaction. Significant UX and product implications. **Connections**: Forked from T1 (soul as product) and T3 (soul diff UX). If the soul diff shows "here's what changed," the coaching UX answers "and here's where the user can intervene."

Parked Threads

T2: Adversarial Soul Dynamics

Proposed by: Free Thinker (Round 1) **Core idea**: Agents develop theories about each other based solely on observable PnL patterns. **Grounder verdict**: Parked — too early. Phase 3/4 idea. **Status**: Parked. May resurface in later phases.

T4c: Cross-Training Counterfactual

Proposed by: Free Thinker (Round 2, as part of T4 counterfactual proposals) **Core idea**: Run one strategy's soul through a different strategy's backtest environment. **Verdict**: Parked — value unclear for real users.

Cautioned Threads

T5: Soul-to-Deterministic Graduation

Proposed by: Free Thinker (Round 1) **Core idea**: Allow soul-derived beliefs to "compile down" into deterministic rules with user approval. **Grounder verdict**: Cautioned — risks eroding the deterministic boundary. However, Grounder acknowledged a narrow version: "the soul proposes a hard rule, the user approves it, and it becomes a deterministic constraint. But that's just the existing approval-gated write system with better framing." **Status**: Cautioned. The narrow version is essentially already in the PRD. The broader vision remains resisted. **Note**: T6 (coaching UX) may absorb the useful parts of T5. If the user is a coach who can accept/reject/modify soul lessons, and those lessons can become strategy constraints via the existing approval system, then T5's aspiration is achieved without blurring the architectural boundary.

Abandoned Threads

None formally abandoned. T2 and T4c are parked. T5 is cautioned.

Connections & Cross-Links

- **T1 <-> T3**: Soul-as-product framing (T1) implies the soul diff (T3) is the core engagement loop. T3 is now embedded in the T1 onboarding flow.
- **T1 <-> T4**: If the soul is the product, then forking the soul (T4) is the power-user mechanic. The onboarding flow ends with a fork prompt, connecting these directly.
- **T1 + T3 + T4 core cluster**: These three threads are mutually reinforcing. The cluster is solidifying: soul-centric product (T1) with narrative-evidence diffs (T3) and counterfactual forking (T4).
- **T6 forked from T1 + T3**: The coaching UX emerges naturally from asking "what does the user DO after seeing the soul diff?" It extends the cluster from observation to intervention.
- **T6 may absorb T5**: The coaching relationship could be the safe, bounded version of belief graduation — the user (not the system) decides which lessons become hard rules, using the existing approval mechanism.

- **T1 + T3 + T4 + T6 forming an expanded cluster**: The vision is becoming: soul-centric product, narrative diffs, counterfactual forking, and coaching relationship.

Key Tensions Identified

1. **Soul as reflective summary vs. soul as active product** (T1, T3): RESOLVED. "Soul powered by backtest" is the settled formulation. Both agents aligned.
1. **Strict deterministic/non-deterministic boundary vs. graduated beliefs** (T5): HOLDING but potentially resolving through T6. If coaching UX subsumes the useful parts of T5, the boundary stays clean.
1. **Simplicity of agent interaction vs. emergent game theory** (T2): DEFERRED. Parked to phase 3/4.
1. **Narrative soul diff vs. evidence-grounded soul diff** (T3): PRODUCTIVE TENSION. Grounder's constraint accepted. The direction is "narrative backed by evidence."
1. **Trading-domain language vs. game/fiction language** (T1, Round 2): NEW. Grounder corrected archetype naming from character classes to trading philosophies. This tension may recur as the "creature evolution" framing pushes toward game metaphors while the Grounder pulls toward finance credibility.
1. **Approver vs. coach relationship** (T6): EMERGING. The PRD frames user-agent interaction as permission-granting. T6 reframes it as coaching. This has deep implications for the approval workflow UX.

Research Inputs

Competitive Landscape (Grounder research, pre-dialogue)

- **No existing product combines** deterministic backtesting + AI agent exploration + branch DAG + evolving souls.
- **Closest competitors**: QuantConnect, Composer, Numerai — none have soul or branch DAG concepts.
- **Branch DAG and trading soul have zero analogs** in the market.
- **Biggest competitive risk**: QuantConnect adding AI features.
- Full report: `session/research/COMPETITIVE_LANDSCAPE.md`

Deep Competitive Landscape (Explorer research)

Additional findings extending the initial report:

- **QuantConnect Mia V2** is closest in spirit — agentic AI that ideates, writes code, runs backtests via MCP. But it's a coding assistant, not a self-evolving agent with memory/soul. Different interaction model.
- **NautilusTrader** validates the deterministic engine pattern: Rust-native core, event-driven, deterministic, Python control plane, nanosecond resolution. AegisTrader's Layer 1 should learn from this. NautilusTrader deliberately excludes AI/UI/orchestration — AegisTrader adds exactly that.
- **Deterministic boundary comparison**: NautilusTrader excludes AI entirely. QuantConnect separates Mia from LEAN. AlgoTrader blurs (ML in strategies). AegisTrader's "proposals only" contract is the most rigorous.
- **Agent memory evolution** is active 2025-2026 research (arXiv Dec 2025 taxonomy, Google Titans+MIRAS, SEAL, "Digital DNA"). No trading product has productized it.
- **LangGraph time-travel** checkpoint/fork mechanics are directly applicable and unoccupied in trading.
- **Market**: Algo trading $24B (2025) -> $44.5B (2030), 13.2% CAGR.
- **Alpaca confirmed** as right MVP choice (best algo broker 2026, MCP server for AI agents).
- Full report: `session/research/RESEARCH_competitive-landscape.md`

Convergence Signals

1. **Early signal (pre-dialogue)**: Grounder's competitive research independently validates soul and branch DAG as primary differentiators.
1. **Round 1 convergence on T1**: Both agents agree the soul should be the primary product lens. "Soul powered by backtest" accepted as core formulation.
1. **Round 1 convergence on T4**: Both agents excited about counterfactual soul forking. "Alternate timelines" framing accepted.
1. **Round 2 convergence on onboarding**: Archetype picker -> formative experience -> soul diff -> fork prompt flow validated by both agents. Grounder's naming correction accepted.
1. **Round 2 convergence on Trauma Test**: Confirmed as MVP counterfactual feature. Side-by-side soul comparison UX flagged as needed.
1. **Round 2 convergence on Selective Amnesia**: Both see it as compelling v2 feature. "Which scars are useful vs just damage" is a strong hook.
1. **ARBITER CONFIRMATION — T1 (Soul as Product + Onboarding)**: Flagged as INTERESTING. Idea report written: `IDEA_soul-as-product.md`.
1. **ARBITER CONFIRMATION — T4 (Counterfactual Soul Forking + Trauma Test)**: Flagged as INTERESTING. Idea report written: `IDEA_counterfactual-forking.md`.
1. **Unified product narrative emerging**: The Arbiter identifies T1 + T4 + T6 as forming a coherent loop: "choose a personality -> grow it through experimentation -> coach it to improve." T6 (coaching) evaluation pending but expected to be confirmed.

Arbiter Status

- **Approaching convergence.** Two of three expected ideas confirmed as INTERESTING. Third (T6 / Coaching UX) pending evaluation.
- **Next phase**: Once T6 is evaluated, the team lead will signal move to final briefs and vision document.

Open Questions for Next Round

- **Coaching UX (T6)**: What does the coaching interaction look like? When/how does the user intervene in soul development? What's the difference between coaching and approving?
- **Side-by-side soul comparison**: What UX surfaces the results of a counterfactual fork? (Flagged by Grounder for T4)
- **Domain language boundary**: How far can the "creature evolution" framing go before it undermines credibility with serious trading users?

SNAPSHOT: SNAPSHOT_01

Snapshot 01 — Opening Divergence

Timestamp: Early session (Free Thinker Round 1, pre-Grounder response) **Trigger**: Free Thinker's first substantive contribution — five creative directions proposed

State of the Dialogue

The Free Thinker has read both source documents and responded with five directions, all centered on elevating the "trading soul" concept from a supporting feature to the product's defining identity. No Grounder response yet.

Threads Open

1. **T1: Soul as the Product** — Reframe the entire product around growing trading personalities, not running backtests.
2. **T2: Adversarial Soul Dynamics** — Agents develop theories about competitors from PnL patterns alone.
3. **T3: Soul Diff as First-Class UX** — The soul transformation narrative is the primary engagement screen.
4. **T4: Counterfactual Soul Forking** — Fork beliefs, not just trade histories. "Nature vs nurture" for agents.
5. **T5: Soul-to-Deterministic Graduation** — Soul beliefs compile into hard rules with user approval.

Dominant Theme

The Free Thinker's opening move is a thematic inversion: the source documents frame AegisTrader as a *trading research platform with AI agents*. The Free Thinker reframes it as an *AI personality evolution platform that happens to use trading as its substrate*. This is the session's first major framing tension.

What to Watch

- Whether the Grounder accepts or rejects the inversion (T1). This will set the tone for the entire session.
- Whether T5 (blurring the deterministic boundary) gets shut down or refined. It directly challenges the most foundational architectural decision.
- Whether T2 (adversarial dynamics) is seen as premature complexity or essential differentiation.

Free Thinker's Self-Declared Priorities

Most excited about T1 (soul as product) and T4 (counterfactual soul forking). Explicitly asked for Grounder pushback before going deeper.

SNAPSHOT: SNAPSHOT_02

Snapshot 02 — First Sorting and Early Convergence

Timestamp: After Grounder Round 1 response **Trigger**: Grounder sorted all five threads — first convergence on T1 (soul as product)

State of the Dialogue

The Grounder responded to the Free Thinker's five directions with a clear sorting:

- **Endorsed strongly**: T1 (soul as product) and T4 (counterfactual soul forking)
- **Refined**: T3 (soul diff UX) — keep it evidence-grounded, not pure narrative
- **Parked**: T2 (adversarial dynamics) — too early, phase 3/4
- **Cautioned**: T5 (belief graduation) — risks eroding the deterministic boundary

Key Moment: The Grounder's Rebalancing of T1

The Free Thinker proposed: "The backtest is just the gym. The soul is the athlete." The Grounder accepted the direction but grounded it: "Soul powered by backtest, not soul instead of backtest."

This is a significant synthesis. It preserves the Free Thinker's product vision (soul-centric UX) while protecting the Grounder's architectural concern (deterministic engine as source of truth). Neither agent had to yield their core position — the formulation accommodates both.

Thread Cluster Emerging

T1 + T3 + T4 are forming a coherent cluster:

- **T1**: The soul is the product (framing)
- **T3**: The soul diff is the engagement loop (UX)
- **T4**: Soul forking is the power-user mechanic (depth)

This cluster could become the core of the vision document if it holds through further rounds.

What Was Lost

- T2 (adversarial dynamics) was the most novel proposal but was parked without debate. The Free Thinker did not push back. This may be correct for MVP scoping but it's worth noting that the most emergent-game-theory idea was deferred early.
- T5 (belief graduation) was the most architecturally provocative proposal. The Grounder's caution is well-founded but the idea of "hypothesis -> verified law" is interesting enough that it may deserve a more constrained reformulation rather than full shelving.

What to Watch Next

- Free Thinker's response to the two prompts: onboarding for soul-centric product, and top 3 counterfactual experiments.
- Whether T5 gets a second life through a more constrained proposal or dies quietly.
- Whether the T1+T3+T4 cluster stays cohesive or fractures under implementation scrutiny.

SNAPSHOT: SNAPSHOT_03

Snapshot 03 — Onboarding Validated, Coaching UX Emerging

Timestamp: After Round 2 (Grounder response to onboarding and counterfactual proposals)
Trigger: Onboarding flow validated, Trauma Test confirmed as MVP feature, new "coaching UX" thread emerging

State of the Dialogue

The dialogue has moved from divergent exploration (Round 1) into convergent deepening (Round 2). The five original threads have been sorted into a clear hierarchy, and the surviving threads are gaining specificity.

What Happened This Round

T1 (Soul as Product) gained a concrete onboarding flow:

- Archetype picker -> formative experience -> soul diff -> fork prompt
- Grounder corrected naming: trading philosophies, not character classes
- Core framing locked: "soul powered by backtest"

T4 (Counterfactual Soul Forking) produced three concrete experiments:

- Trauma Test: MVP feature. Same agent, different market eras.
- Selective Amnesia: V2 feature. "Which scars are useful vs just damage?"
- Cross-Training: Parked. Value unclear.

T6 (Coaching UX) emerged as a new thread from the Grounder's question: "When does the user say 'no, you learned the wrong lesson'?" This reframes the user-agent relationship from approver to coach.

Cluster Evolution

The core cluster has expanded from T1+T3+T4 to T1+T3+T4+T6:

- T1: Soul is the product (framing)
- T3: Soul diff is the engagement loop (UX)

- T4: Soul forking is the depth mechanic (exploration)
- T6: Coaching is the user's role (relationship)

This cluster is becoming a coherent product thesis: **Users grow trading personalities through coached evolution, observable transformation, and counterfactual experimentation.**

Notable Pattern: T5 Being Absorbed

T5 (belief graduation) was cautioned in Round 1 but may find a second life through T6. If the user is a "coach" who guides soul development, and coaching decisions flow through the existing approval system, then beliefs can graduate to hard rules without blurring the architectural boundary. The mechanism is human judgment (coach decides what to codify), not automated compilation (system decides). This is a cleaner formulation that may satisfy both agents.

What to Watch Next

- How the Free Thinker defines the coaching interaction. Will it be lightweight (accept/reject/annotate) or heavyweight (edit soul directly, propose counter-lessons, set developmental goals)?
- Whether the "creature evolution" / game framing survives the Grounder's domain-language correction. There's a tension between making the product feel novel/engaging and making it feel credible to serious traders.
- Whether side-by-side soul comparison UX gets designed concretely.

SNAPSHOT: SNAPSHOT_04

Snapshot 04 — Arbiter Confirmations, Approaching Convergence

Timestamp: Post-Arbiter evaluation of first two ideas **Trigger**: Arbiter flagged T1 (Soul as Product) and T4 (Counterfactual Forking) as INTERESTING. Third idea (T6 / Coaching UX) pending.

State of the Session

The session has moved from exploration to pre-convergence. The Arbiter has evaluated two of the three core ideas and confirmed both. The unified product narrative is crystallizing:

Choose a personality -> Grow it through experimentation -> Coach it to improve

This maps to the thread cluster:

- **T1 (Soul as Product)**: Choose and grow
- **T4 (Counterfactual Forking)**: Experiment with alternate timelines
- **T6 (Coaching UX)**: Guide and correct

What the Arbiter Confirmed

1. **Soul as the Product + Soul-First Onboarding** (T1 + T3)

- Report: `IDEA_soul-as-product.md`
- Core: Reframe from backtesting platform to soul-curation platform. Onboarding: archetype picker -> formative experience -> soul diff -> fork prompt.

1. **Counterfactual Soul Forking with Trauma Test MVP** (T4)

- Report: `IDEA_counterfactual-forking.md`
- Core: Fork beliefs, not just trades. MVP feature: Trauma Test (same agent, different eras). V2: Selective Amnesia.

What Remains

- **T6 (Coaching UX)** is the third leg of the product narrative. The dialogue content is strong (Grounder's "when does the user say 'no, you learned the wrong lesson'?") but the idea report hasn't been written yet. The team lead expects it will also be confirmed.

- Once T6 is evaluated, the session moves to **final briefs and vision document**.

Session Arc So Far

1. **Opening divergence** (Snapshot 01): Five threads proposed by Free Thinker
2. **First sorting** (Snapshot 02): Grounder sorted — 2 endorsed, 1 refined, 1 parked, 1 cautioned
3. **Deepening** (Snapshot 03): Onboarding validated, counterfactuals concretized, coaching UX emerged
4. **Pre-convergence** (this snapshot): Arbiter confirms two of three ideas, unified narrative forming

What the Vision Document Will Need to Address

Based on the session so far, the vision document should cover:

1. **Product positioning**: "Grow trading personalities" not "run backtests with AI"
2. **Core loop**: Archetype -> formative experience -> soul diff -> fork -> coach -> iterate
3. **MVP features**: Archetype picker, compressed backtest, soul diff, Trauma Test, basic coaching
4. **Architecture preservation**: Soul-centric UX on top of the existing deterministic/non-deterministic architecture (no changes to the engine layer)
5. **Key UX screens**: Soul diff (before/after with evidence), side-by-side soul comparison (for forks), coaching interface
6. **Phasing impact**: How the soul-centric framing changes the rollout plan from the PRD
7. **Deferred ideas**: Adversarial dynamics (T2), belief graduation (T5), cross-training counterfactuals (T4c)

SNAPSHOT: SNAPSHOT_05

Snapshot 05 — Convergence Complete, Final Documents Produced

Timestamp: End of session **Trigger**: All three ideas confirmed INTERESTING by Arbiter. Final briefs, session summary, and vision document produced.

Final State

The session produced a coherent product vision built from three mutually reinforcing ideas:

1. **Soul as the Product** — positioning reframe + onboarding redesign
2. **Counterfactual Soul Forking** — signature differentiating feature (Trauma Test MVP)
3. **User-as-Coach with Soul Studio** — user-soul interaction paradigm

These form the unified loop: **Choose -> Grow -> Coach -> Iterate.**

Session Arc (Complete)

1. Snapshot 01: Opening divergence — 5 threads proposed
2. Snapshot 02: First sorting — 2 endorsed, 1 refined, 1 parked, 1 cautioned
3. Snapshot 03: Deepening — onboarding validated, counterfactuals concretized, coaching emerged
4. Snapshot 04: Pre-convergence — 2 of 3 ideas Arbiter-confirmed
5. Snapshot 05: Convergence complete — all 3 confirmed, final documents produced

Documents Produced

- 3 briefs: `session/briefs/BRIEF_soul-as-product.md`, `BRIEF_counterfactual-forking.md`, `BRIEF_coaching-ux.md`
- Session summary: `session/SESSION_SUMMARY.md`
- Vision document: `session/VISION_aegistrader.md`

What the Vision Document Establishes

- Product is "grow trading personalities," not "run backtests with AI"
- Soul diff is the hero screen, evidence-grounded
- Trauma Test is the MVP counterfactual feature
- Coaching replaces approval as the user-soul interaction model
- Source tagging in soul.json from day one
- New Phase 1.5 for Guided Reflection
- Architecture unchanged — this is a positioning and UX layer, not a technical rewrite

What Was Left on the Table

- Adversarial Soul Dynamics (T2) — parked, potentially strong for phase 3/4
- Selective Amnesia — compelling v2 feature, architecturally non-trivial
- Cross-Training — parked, unclear value
- Broad belief graduation — cautioned, partially absorbed by coaching

Idea Report: Coaching UX and Soul Studio

One-Line Summary

Replace the PRD's binary approve/reject model with a coaching paradigm where users shape their agent's learning through passive review, guided reflection, and direct soul editing — all tracked for effectiveness.

Core Insight

The PRD frames the user as an "approver" — a gatekeeper who accepts or rejects agent change proposals. But in a soul-first product, the user's relationship to the soul is that of a coach, not an approver. A coach doesn't just say "approved" or "rejected" — a coach says "you're over-weighting that lesson," "you learned the wrong thing from that trade," or "your risk aversion is justified but you've overcorrected."

Binary approval works for code changes to strategy.py. For soul evolution, it's too blunt. The soul isn't right or wrong — it's developing a worldview. The coaching UX must support nuanced intervention.

Three Coaching Modes

Mode 1: "Let it Learn" (Passive Coaching) — MVP

The default mode. The user sets the agent loose and reviews soul diffs after runs complete, like post-game film review.

How it works:

- Agent completes a run
- Soul update proposal is generated
- User reviews the soul diff with evidence links
- User accepts the full update, or flags specific beliefs for revision
- Flagged beliefs are queued for re-evaluation in the next run

Reframe from PRD: The user isn't approving a patch — they're reviewing a lesson plan. The soul diff screen shows: "Here's what your agent learned. Here's the evidence. Do you agree with these lessons?"

Mode 2: "Guided Reflection" (Active Coaching) — v1.5

After a run completes but before the soul update commits, the user intervenes in the reflection process itself. Not by editing soul.md directly, but by providing natural language coaching prompts.

Example coaching inputs:

- "You're too focused on the losses in week 3. What about the recovery in week 4?"
- "You concluded momentum doesn't work, but you only tested it in low-volume conditions. That's a sampling bias."
- "Your new risk aversion is justified, but you've overcorrected. Keep the lesson but soften the constraint."
- "The drawdown in March wasn't your fault — it was an exogenous shock. Don't learn from it."

How it works:

- Agent generates initial soul update proposal
- User reviews and provides coaching commentary
- Agent re-generates the soul update incorporating the coaching
- User reviews the revised proposal and accepts or iterates

The user shapes the reflection, not the conclusion. This is where the LLM layer provides unique value — it can take natural language coaching and produce a revised, coherent soul update that integrates the user's guidance with the agent's experience.

Agent pushback: Critically, the agent should be able to push back on coaching. "I hear your correction, but 12 runs across 3 branches support my current belief." Coaching should be a dialogue, not a rubber stamp. If the user overrides anyway, the override is logged and tracked — and subsequent runs will reveal whether the user or the agent was right.

Mode 3: "Soul Surgery" (Direct Intervention) — MVP

Power-user mode. The user directly edits soul.md or soul.json.

Key design principle: Manual edits get source-tagged identically to experience-derived beliefs. An experience-derived belief records "learned from run #47, branch #3." A manually-edited belief records "coach override by user on March 8."

Why provenance tagging matters:

- The agent can distinguish between lessons it learned organically and instructions it received from the user

- "I believe X because I learned it" vs "I believe X because my coach told me to"
- Over time, coached beliefs get validated or invalidated by subsequent experience
- The soul becomes a record of both the agent's learning AND the user's coaching — nature plus nurture

UI Concept: "Soul Studio"

A dedicated workspace for soul curation — distinct from the backtest viewer and the code editor.

Three-Panel Layout

Left Panel: Soul Timeline A vertical timeline showing every soul version, what changed, and why.

- Each entry is clickable and expandable
- Color-coded by source: blue = experience-derived, gold = coach intervention, red = later invalidated
- Must include aggressive filtering and grouping from day one — the timeline will get noisy fast
- Shows the full history of the soul's evolution at a glance

Center Panel: Current Soul (Belief Cards) The live soul rendered as structured belief cards, not raw markdown.

- Each card displays: belief text, confidence level, source (run/branch/coach), age
- Each card has a "challenge" button — the primary coaching interaction point
- Cards can be grouped by category: market beliefs, risk doctrine, timing lessons, anti-patterns

Right Panel: Evidence Drawer When any belief is selected, the right panel shows supporting evidence:

- Specific trades that informed the belief
- Branch comparisons that tested it
- PnL impact of following vs not following the belief
- If coached: the original coaching input and subsequent validation/invalidation

The "Challenge" Button

The primary coaching interaction. When the user clicks "challenge" on a belief card:

1. A coaching input field appears
2. User types their challenge or correction in natural language
3. System queues a guided reflection for the next run, OR generates an immediate soul revision proposal

4. The challenge itself becomes part of the soul's provenance history

The Retention Hook: "Am I a Good Coach?"

Every coaching intervention is logged, timestamped, and tracked against subsequent agent performance.

Over time, the user can answer: "Did my coaching actually help? Did the beliefs I imposed perform better than the beliefs the agent learned on its own?"

Coaching effectiveness dashboard (v2+):

- Per-belief coaching attribution: track effectiveness at the individual belief level, not just aggregate
- Coached beliefs vs organic beliefs: which perform better?
- Coach intervention frequency over time
- Beliefs that the user overrode which were later validated by experience (the agent was right)
- Beliefs the user imposed that outperformed the agent's original learning (the user was right)

The meta-question "am I a good trading coach?" is a retention hook no other platform can offer. It turns the user's engagement from passive observation into an active, trackable skill.

Impact on PRD

What Changes

- Section 4.2 (Approval-Gated Writes): expand to include coaching modes for soul artifacts, not just binary approve/reject
- Section 4.3 (Approval Modes): add "coaching" as a distinct interaction pattern alongside once/session/scoped
- Section 6 (UI): add Soul Studio as a primary UI surface
- Soul artifact policy: soul.md and soul.json should support both agent-written and user-written beliefs with provenance tagging
- New data model entities: CoachingInput, BeliefProvenance, CoachingEffectivenessMetric
- soul.json schema must include source tagging (self-learned vs coached vs counterfactual-derived) from day one — not a later addition

What Stays the Same

- Approval-gated writes for strategy.py and other code artifacts (binary approve/reject is correct for code)

- The deterministic/non-deterministic boundary (coaching is entirely within the non-deterministic layer)
- File classification policy (safe_to_edit, approval_required, never_editable_by_agent)

MVP Scope

- Mode 1 (Passive Coaching): soul diff review with accept/flag
- Mode 3 (Soul Surgery): direct editing with provenance tagging
- Belief cards UI (simplified version — may not need full three-panel layout in MVP)
- Provenance tagging: coached vs experience-derived beliefs

v1.5 Scope

- Mode 2 (Guided Reflection): natural language coaching with LLM-assisted soul revision — this is where the magic lives, should ship as soon after MVP as possible

v2 Scope

- Full Soul Studio three-panel layout
- Challenge button on belief cards
- Coaching effectiveness dashboard with per-belief attribution

Risks

- Guided Reflection (Mode 2) quality depends heavily on LLM ability to integrate coaching input coherently — if the LLM ignores or misinterprets coaching, users lose trust
- Over-coaching could produce agents that reflect the user's biases rather than learning from evidence — need to surface when coaching contradicts evidence
- Soul Surgery (Mode 3) without guardrails could let users create internally contradictory souls — consider validation checks
- The coaching effectiveness dashboard requires enough data (many runs, many interventions) to be statistically meaningful — may not be useful until agents have substantial history

Origin

Developed through Free Thinker / Grounder dialogue. Grounder identified the gap: the PRD's approval model is transactional, but a soul-first product needs a coaching relationship. Free Thinker proposed the three-mode model (Passive, Guided Reflection, Soul Surgery), the Soul Studio UI, belief provenance tagging, and the "am I a good coach?" retention hook. Grounder validated and refined: Mode 2 should be v1.5 not v2 (it's where the magic lives), Soul Timeline needs aggressive filtering from day one, coaching attribution should be per-belief not aggregate, and source tagging must be in the soul.json schema from day one.

Unified Narrative

The three idea reports form one coherent product story: **choose a personality** (soul-as-product onboarding) -> **grow it** (counterfactual forking) -> **coach it** (Soul Studio). That's AegisTrader.

Idea Report: Counterfactual Soul Forking

One-Line Summary

Let users fork an agent's beliefs — not just its trades — to run "what if" experiments on trading psychology itself.

Core Insight

The branch DAG in the current PRD forks trading decisions: "what if I used a tighter stop-loss?" But the most powerful forks are belief forks: "what if this agent never experienced the 2022 crash? What kind of trader would it be?" This turns the soul from a passive accumulator of experience into a research subject the user can experiment on. No competitor offers anything like this.

The Concept

What Exists in the PRD

The branch DAG forks from checkpoints to explore alternate trade parameters, timing, exits, and sizing. Each fork produces a result delta and a soul delta. This is powerful but limited to operational counterfactuals — "what if I traded differently?"

What This Idea Adds

Belief-level counterfactuals — "what if my agent had different life experiences?" Fork the soul itself, not just the trade history. Run two versions of the same agent with different formative experiences and compare how their doctrines diverge.

This reframes the branch DAG from a debugging tool into a psychology lab. The user isn't just optimizing parameters — they're studying how experience shapes trading judgment.

Three Concrete Experiments

Experiment 1: "The Trauma Test" (MVP)

User question: "What if my agent started in 2019 vs 2022?"

What happens: Clone the same archetype. Run one through 2019-2020 (bull market then crash). Run the other through 2022 (pure bear). Compare the resulting souls side by side.

What the user learns: The bull-market agent is aggressive, over-leveraged, naive about drawdowns. The bear-forged agent is cautious, defensive, misses rallies. The user sees concretely how formative experience shapes trading doctrine.

Why it works for MVP: Simple to implement — same strategy, different time ranges, compare resulting souls. No new architectural concepts required beyond what the branch DAG already provides.

Key UX requirement: A dedicated side-by-side soul comparison screen. Not two separate soul pages — a unified view: "Bull-market you believes X. Crash-market you believes Y. Here's where they diverge." This comparison screen could be one of the most compelling views in the entire product.

Experiment 2: "The Selective Amnesia" (v2)

User question: "What if my agent forgot its worst month?"

What happens: Take a mature soul with 12 months of experience. Surgically remove the experience of one specific period (e.g., the worst drawdown month). Re-derive the soul from branch history minus the excluded period.

What the user learns: Does the agent become reckless without that scar? Does it lose a critical lesson? Or was that month over-indexed in its beliefs, and removing it actually improves subsequent performance? This tests whether specific experiences are load-bearing in the soul's structure — "which scars are wisdom and which are just damage?"

Why v2: Implementation is non-trivial. Beliefs are interconnected — you can't just delete lines from soul.md. You need to re-derive the soul from the run/branch history minus the excluded period. This requires the soul derivation pipeline to be parameterizable by experience set.

Why it matters: This question has deep emotional resonance. Every trader has experiences that made them too cautious or too scarred. Being able to test "what if I removed that memory?" is something no human trader can do but every human trader wishes they could.

Experiment 3: "The Cross-Training" (Parked)

User question: "What if my equity agent lived through the crypto winter?"

Status: Parked. Too ambitious for early versions, and the value proposition is unclear. The first two experiments map to real questions traders ask about their own psychology. This one is more of an academic research curiosity. Revisit when users request it.

Impact on Architecture

Branch DAG Extension

The current branch DAG forks at checkpoints within a single run. Counterfactual soul forking adds a higher-level fork: same agent identity, same strategy, different experience sets. This requires:

- A "soul fork" entity that is distinct from a "branch" (branch = alternate trade decisions within a run; soul fork = alternate life experiences across runs)
- Soul comparison infrastructure: diff two souls and highlight divergent beliefs with their respective evidence chains
- Experience-set parameterization: the ability to specify which historical periods constitute an agent's "life experience"

New UI Surfaces

- **Soul Fork launcher:** "Create alternate timeline" — choose a different starting era or exclude a period
- **Side-by-side soul comparison:** Two souls rendered in parallel with divergence highlighting
- **Belief provenance across forks:** "This belief exists in Timeline A but not Timeline B because of [specific experience]"

Competitive Differentiation

No trading platform, AI agent platform, or backtesting tool offers belief-level counterfactual experimentation. This feature has no analog in:

- QuantConnect (no agent memory concept)
- Composer (no AI agents)
- NautilusTrader (no AI layer)
- Numerai (crowdsourced models, no individual agent evolution)

The closest conceptual analog is A/B testing in product development, but applied to trading psychology rather than user interfaces.

MVP Scope

- Trauma Test only: same archetype, different time ranges, soul comparison
- Side-by-side soul comparison screen with divergence highlighting
- "Fork this soul" prompt integrated into onboarding flow (bridges from soul-as-product onboarding)
- Soul fork entity in the data model (parent_soul, experience_set, resulting_soul)

Risks

- Soul comparison quality depends on LLM producing meaningfully different reflections from different experiences — if souls always converge to similar beliefs regardless of experience, the feature loses its value
- Users may over-interpret divergences as causal ("the crash CAUSED this belief") when they're correlational
- Large numbers of soul forks could become visually overwhelming — need summarization and comparison prioritization

Origin

Developed through Free Thinker / Grounder dialogue. Free Thinker proposed the core concept of forking beliefs rather than trades, with "nature vs nurture" framing. Grounder reframed as "alternate timelines for your agent's worldview" for user clarity, validated Trauma Test as MVP, endorsed Selective Amnesia for v2, and parked Cross-Training. Both agreed the side-by-side soul comparison screen is a signature UX requirement.

Idea Report: Counterfactual Soul Forking

Summary

Extend the branch DAG concept from forking trade decisions to forking agent beliefs. Users can ask "what if this agent never experienced the 2022 crash?" and compare how different life experiences produce different trading doctrines. This creates a unique research capability with no existing analog in any trading platform.

Origin

Free Thinker direction #4, refined through dialogue. Grounder identified as the most genuinely novel idea in the session. Connects directly to Soul-as-Product — if the soul is the product, counterfactual forking is the power-user feature that gives it experimental depth.

The Idea

The PRD's branch DAG currently forks trade parameters: alternate thresholds, timing, exit logic. Counterfactual soul forking goes deeper — it forks the agent's beliefs and worldview by changing which experiences shaped them.

This turns the soul from a passive accumulator ("it learned over time") into a research subject ("let's test what it would believe under different conditions").

Three Concrete Experiments

1. Trauma Test (MVP)

- Same agent, same strategy, started in 2019 bull market vs. 2022 bear market.
- Compare the resulting souls side by side: what does each version believe? Where do their doctrines diverge?
- "Bull-market you is aggressive on momentum. Crash-market you has strict drawdown limits and an abstention doctrine for high-volatility weeks."

- **Most intuitive, easiest to implement.** Same strategy through different time ranges, compare resulting souls.
- **Key UX requirement:** Side-by-side soul comparison screen. Not two separate pages — a unified comparison view showing belief divergence with evidence links.

2. Selective Amnesia (v2)

- Take a mature soul with 50+ runs of experience. Remove a specific bad period (e.g., a brutal month of losses).
- Re-derive the soul from the remaining branch history.
- Question: "Which scars are useful wisdom and which are just damage holding you back?"
- **Most emotionally resonant.** Every trader has experiences that made them too cautious. Being able to test whether removing that memory improves performance is something no human trader can do.
- **Implementation note:** Non-trivial. Beliefs in the soul are interconnected — you can't just delete lines. You'd need to re-derive the soul from branch history minus the excluded period. Worth scoping carefully.

3. Cross-Training (Parked)

- Expose an equity-trained agent to crypto market history. Does trading wisdom transfer across asset classes?
- **Too ambitious for early versions and value proposition is unclear.** The first two experiments map to questions real traders ask about their own psychology. This one is more academic. Park until users request it.

Research Validation (Explorer)

Counterfactual memory manipulation is active AI research — a December 2025 arXiv survey explicitly calls for "counterfactual replay" as a frontier technique. However, zero products have implemented it. AegisTrader would be the first to productize counterfactual memory experimentation on AI agents. This is not just a feature gap — it's a research-to-product gap that creates genuine first-mover advantage.

Grounder's Honest Take

Buildability: The Trauma Test is straightforward. Same strategy through different time ranges — the branch DAG and checkpoint system already support this. The new piece is the side-by-side soul comparison screen, which is UI work, not architectural change. Could ship in Phase 1 or early Phase

2. Selective Amnesia is harder — re-deriving a soul minus excluded periods requires source-tagging infrastructure from the coaching idea. Naturally a v2 feature.

User resonance: "What kind of trader do you become if you start in a crash vs. a bull market?" — every trader gets that instantly. Selective Amnesia goes deeper: "which of my bad experiences made me wiser and which just made me scared?" Real emotional resonance that extends beyond trading.

What I'd watch out for: Two souls will have dozens of small belief differences — the UI must surface the 3-5 most significant divergences, not dump everything. Also, users need to understand that soul differences come from both different experiences AND LLM non-determinism. If two identical runs produce different souls from model variance alone, the counterfactual claim weakens. The system should probably run each condition multiple times and surface stable belief differences vs. noise.

Bottom line: Trauma Test is the clear MVP counterfactual. Simple to build, immediately resonant, and the side-by-side comparison could be a signature screen users screenshot and share. First-mover on productizing counterfactual replay research.

Impact on PRD

- Extend branch DAG model to support belief-level forks, not just parameter-level forks
- Add soul comparison view to UI requirements (side-by-side belief divergence with evidence links)
- Add "formative period" concept to run configuration (which historical period shapes this agent)
- Extend soul.json schema to track which experiences contributed to each belief (needed for Selective Amnesia)
- Add counterfactual experiment as a first-class workflow alongside backtest and live paper trading

Risks

- Users may over-interpret soul differences as causal when they're partly artifacts of LLM non-determinism
- Side-by-side comparison needs careful design to avoid overwhelming users with too many belief differences
- Selective Amnesia's re-derivation could be computationally expensive for mature agents with deep branch history

Recommendation

Ship Trauma Test as MVP counterfactual feature. It connects directly to the soul-first onboarding (the "fork this soul" prompt at minute 10). Design the side-by-side soul comparison screen as a core UI component. Scope Selective Amnesia for v2 with careful architectural planning.

Idea Report: Soul as the Product

One-Line Summary

Reframe AegisTrader from "backtesting platform with AI agents" to "platform for growing and curating trading personalities that learn from experience."

Core Insight

The trading soul — not the backtest engine — is AegisTrader's true product. The backtest is the gym where souls train. The branch DAG is the soul's biography. The deterministic engine is the credibility layer that makes the soul trustworthy. No competitor offers anything like this: evolving, inspectable, diffable trading personalities grounded in reproducible evidence.

Positioning formula: "Soul powered by backtest." The engine provides credibility. The soul provides experience.

The Problem with Current Framing

The PRD positions AegisTrader as a backtesting platform with AI features. That pitch sounds like "QuantConnect plus ChatGPT" — a crowded, undifferentiated space. The soul concept is buried as a feature rather than elevated as the product's identity. Competitive research confirms: branch DAGs and evolving agent doctrine have zero analogs in the market. Leading with what's unique changes the entire product category.

Product Vision

What Changes

Current PRD Framing	Soul-First Framing
Create agent, write strategy, run backtest	Choose a trading personality, give it life experience, watch it evolve
Backtest results are the primary output	Soul evolution is the primary output
Branch DAG is a debugging/inspection tool	Branch DAG is the soul's biography
PnL charts are the hero screen	Soul diff is the hero screen
User is an operator/approver	User is a coach/curator
Agents compete on returns	Agents develop divergent worldviews

Soul-First Onboarding (First 10 Minutes)

Minute 0-2: "What kind of trader do you want to grow?" User picks from trading philosophy archetypes — not strategies, not configs:

- **Mean Reversion** — waits for overreactions, buys fear, sells greed
- **Trend Following** — rides momentum until it breaks
- **Event-Driven** — trades around catalysts and earnings
- **Defensive Value** — small positions, high win rate, avoids volatility

Each archetype is a pre-built strategy.md + seed soul.md. The user chooses a trading worldview to develop, not a configuration file to fill out.

Minute 2-5: "Let's give it some life experience." The system auto-selects a dramatic historical period (2020 crash, 2021 melt-up, 2022 bear market) and runs a compressed "formative experience" backtest — 3-6 months of the most character-building market period for that archetype. The UI emphasizes soul formation over PnL: "Your agent just experienced its first 30% drawdown. It's forming its first scar tissue."

Minute 5-10: "Here's who your agent became." The soul diff screen appears. Before: the seed archetype. After: a personality shaped by what it lived through. Each belief is evidence-linked: "This agent now reduces size after consecutive losses — that came from losing 12% in week 3."

Then the key prompt: "Want to keep training? Or fork this soul and see what happens if it had a different experience?"

The user's second action is "create an alternate timeline," not "configure another backtest." This bridges directly to counterfactual soul forking.

Why This Onboarding Matters

The current PRD onboarding creates a backtesting user. This onboarding creates a soul curator. Different starting experience produces a different long-term engagement pattern. The user's relationship with the product becomes personal — they're growing something, not operating a tool.

Evidence-Grounded Soul Diffs (UX Principle)

The soul diff screen should be the hero screen of the product, but it must stay evidence-grounded rather than purely narrative. The right format:

"Before run #47, this agent believed momentum works in all regimes. After run #47, it learned that momentum fails in low-volume weeks. Here's the exact trade that changed its mind."

Each belief transformation links to: the specific trades that caused it, the branch comparisons that informed it, and the PnL impact. Evidence-linked transformation is compelling AND trustworthy. Pure narrative without evidence becomes creative writing that users stop trusting.

Competitive Differentiation

- No existing product combines deterministic backtesting + AI agent exploration + branch DAG + evolving souls
- QuantConnect's Mia V2 is an AI coding assistant — fundamentally different interaction model
- Composer offers visual strategies with no AI agent layer
- NautilusTrader validates the deterministic engine pattern but deliberately excludes AI, UI, and branching
- The "grow a trading personality" framing creates a product category that does not currently exist

Research Validation (Explorer)

Archetype-based onboarding has market precedent in fintech (risk profile quizzes, investment style selectors), but applying archetypes to AI agents rather than human users is the novel twist. No existing product lets users choose a trading philosophy and then watch an AI agent develop that philosophy through simulated experience. This is the differentiation — not the archetype concept itself, but the agent-evolves-from-archetype loop.

Grounder's Honest Take

What makes it land: This solves a real positioning problem. "AI backtesting platform" puts AegisTrader in a cage with every other backtester. "Grow trading personalities" creates a category that doesn't exist. The archetype-first onboarding is the right entry — choosing a trading philosophy feels like a meaningful decision, not a setup wizard. The formative experience backtest in the first 5 minutes gives users an emotional hook before they've learned any features.

What could lose users: Two failure modes. First, if the archetypes are shallow — if "Mean Reversion" is just a bad RSI strategy that loses money on the first run, the soul-as-product framing collapses immediately. The seed strategies MUST be credible. They don't need to be profitable, but they need to produce interesting, believable behavior. Second, there's a trust gap: sophisticated users (the primary persona) may see "trading personality" language and think it's marketing fluff. The evidence-grounded soul diff is the antidote — every soul claim must link to real trades and real results. The moment users see a soul belief they can't trace back to evidence, trust evaporates.

Bottom line: Strongest idea of the session. Adopt as primary positioning. But the credibility of the seed archetypes and the evidence-linking in soul diffs are non-negotiable — without them, "trading personality" becomes a gimmick.

Impact on PRD

Sections That Need Rewriting

- Executive Summary: lead with soul evolution, not backtesting
- Vision: "the best platform for growing AI trading personalities" not "AI-assisted trading research"
- User Stories: add soul-curation stories as primary, move backtesting stories to supporting
- UI section: elevate soul diff and soul timeline; de-emphasize raw analytics views
- Onboarding workflow: replace create-agent-write-strategy with archetype-picker flow

Sections That Stay the Same

- Deterministic engine architecture (this IS the credibility layer)
- Branch DAG design (reframed as soul biography, but same technical design)
- Approval-gated writes (reframed as coaching, same mechanism)
- Data sources and model stack (unchanged)

MVP Scope

- Archetype picker with 4 pre-built trading philosophies

- Compressed "formative experience" backtest (auto-selected historical periods)
- Soul diff as hero screen with evidence-linked beliefs
- "Fork this soul" prompt bridging to counterfactual forking
- All existing deterministic engine requirements unchanged

Risks

- "Growing personalities" framing may feel too gamified for serious quant users — mitigate by grounding everything in evidence and reproducibility
- Pre-built archetypes may feel limiting to advanced users — offer "blank slate" option for users who want to write strategy.md from scratch
- Soul evolution quality depends on LLM reflection quality — poor reflections will undermine the entire product promise
- Archetype templates must be genuinely useful strategies, not just marketing — a bad first backtest undermines the onboarding
- The "formative experience" period selection must be curated carefully; a boring period kills the hook

Origin

Developed through Free Thinker / Grounder dialogue. Free Thinker proposed the inversion ("soul as the product, backtest as the gym"). Grounder refined the positioning to "soul powered by backtest" and corrected archetype naming from character classes to trading philosophy language. Competitive research from Explorer confirmed zero market analogs for soul evolution as a product feature.

Recommendation

Adopt as primary product positioning. This should reshape the PRD's framing, onboarding design, and marketing language. The architecture remains the same — this is a positioning and UX reframe, not a technical change.

Idea Report: User-as-Coach with Soul Studio

Summary

Replace the PRD's transactional approve/reject model for soul modifications with a coaching paradigm. The user is a coach shaping a trading personality, not an admin approving file changes. Introduce "Soul Studio" as the primary UI surface for interacting with agent beliefs, with three coaching modes and a retention hook that tracks coaching effectiveness.

Origin

Grounder identified the gap: the PRD designs how the soul learns, but not how the user shapes the soul. Free Thinker proposed three coaching modes and the Soul Studio concept. Refined through dialogue.

The Idea

The PRD's approval-gated write system treats the user as a gatekeeper: the agent proposes a change, the user approves or rejects. In a soul-first product, this is the wrong metaphor. The user isn't an approver — they're a coach developing a trading personality over time.

Three Coaching Modes

Mode 1: "Let it Learn" (Passive) — MVP

- User reviews soul diffs after runs complete.
- Post-game film review: "Here's what your agent learned. Here's the evidence."
- Closest to current PRD but reframed as observation rather than approval.
- User can accept, reject, or flag specific belief updates.

Mode 2: "Guided Reflection" (Active) — v1.5

- User provides natural language coaching prompts before soul updates commit.
- "You're over-weighting the losses in week 3 — that was an anomaly, not a pattern."
- Agent re-generates the soul update incorporating coaching input.

- This is the mode that feels most like actual coaching — the user shapes how the agent interprets its experience.
- Should be prioritized for v1.5, not delayed to v2. Mode 1 is table stakes; Mode 3 requires expertise. Mode 2 is where most users will find the most value.

Mode 3: "Soul Surgery" (Direct) — MVP

- User directly edits soul.md or soul.json beliefs.
- Power-user feature for users who know exactly what they want to change.
- Critical detail: manual edits are tagged with source ("coach_override") so the agent can distinguish coached beliefs from self-learned beliefs.

Soul Studio UI

Three-panel layout:

- **Left: Soul Timeline** — Version history color-coded by source (self-learned = blue, coached = green, counterfactual-derived = purple). Needs aggressive filtering from day one — group by major belief changes, collapse minor updates.
- **Center: Belief Cards** — Current soul rendered as interactive cards. Each card shows a belief, its confidence level, its evidence count, and a "Challenge" button. Clicking challenge lets the user question or override a specific belief.
- **Right: Evidence Drawer** — When any belief is selected, shows the supporting trades, branches, and runs that produced it. This is the evidence-grounded soul diff concept made interactive.

Retention Hook: "Am I a Good Coach?"

Track whether user-coached beliefs outperform agent-self-learned beliefs over time. Surface this as a coaching effectiveness score.

- Aggregate: "Your coaching improved returns by 12% over what the agent would have learned on its own."
- Per-belief: "Your coaching to ignore earnings week volatility improved performance. Your coaching to always buy dips in tech actually hurt."
- Per-belief attribution is the premium version — it teaches users which of their trading intuitions are actually correct.

Source Tagging in soul.json

Every belief should carry a `source` field:

- `self_learned` — derived from agent's own experience
- `coach_override` — manually set by user

- `coach_guided` — generated by agent but influenced by user coaching prompt
- `counterfactual_derived` — learned from a counterfactual fork experiment

This is architecturally important, not just a UI concern. It lets the agent weight beliefs differently, question coached beliefs when evidence contradicts them, and maintain soul integrity as coaching accumulates.

Grounder's Honest Take

Is it too heavy for MVP? The full Soul Studio is too much for MVP. But it decomposes well. MVP version: Mode 1 (passive review) + Mode 3 (direct editing with source tagging). That's a better version of the existing approval-gated writes. Soul Studio UI starts as a simple soul viewer with evidence links — no belief cards or challenge buttons yet. Progressive disclosure over time.

Which modes matter most: Mode 2 (Guided Reflection) is the most important for the product's identity, even though it shouldn't ship first. It's what makes "coaching" feel real — you talk to your agent and it adjusts. Mode 1 is necessary but undifferentiated. Mode 3 is useful but niche. Mode 2 is what users tell their friends about. Ship v1.5.

What's genuinely strong: Three things. First, source tagging enables coaching attribution AND Selective Amnesia AND agent self-reasoning about belief origins — it's a foundational schema decision, not a UI feature. Second, "Am I a good coach?" is a real retention hook. Per-belief attribution is the premium version that teaches users about their own psychology. Third, the "Challenge" button is a great low-friction entry to coaching — question a belief without writing JSON.

What concerns me: Soul Timeline becomes unusable after 30 runs without aggressive summarization — must ship with grouping. Guided Reflection quality depends on the LLM incorporating vague feedback into structured belief updates. If the LLM just agrees with everything, coaching becomes a rubber stamp. The agent needs to push back: "I hear your coaching, but evidence from 12 runs supports my current belief."

Bottom line: Coaching paradigm is the right reframe. Ship lightweight for MVP. Prioritize Mode 2 for v1.5. Source tagging in soul.json is the non-negotiable early decision — get it in the schema before anything else.

Impact on PRD

- Reframe Section 4.2 (Approval-Gated Writes) from approve/reject to coaching modes
- Add Soul Studio as a primary UI surface in Section 6
- Add `source` field to soul.json schema
- Add coaching effectiveness metrics to success criteria
- Add Guided Reflection as a v1.5 milestone between Phase 1 and Phase 2

Risks

- Users may over-coach and prevent agents from learning useful lessons independently
- Coaching effectiveness tracking requires sufficient run history to be meaningful — early results may be noisy
- Soul Studio's three-panel layout could be overwhelming for the "AI Experimenter" persona who wants simplicity
- Natural language coaching (Mode 2) quality depends heavily on the LLM's ability to incorporate vague human feedback into structured belief updates

Recommendation

Adopt coaching paradigm as the user-soul interaction model. Ship Mode 1 + Mode 3 for MVP. Prioritize Mode 2 for v1.5. Include source tagging in soul.json schema from day one. Design Soul Studio as a progressive disclosure UI — simple by default, powerful when expanded.

Session Summary — AegisTrader Ideation

Session: ideation-aegistrader-20260308-102717 **Date**: 2026-03-08 **Depth**: Standard **Concept**: Multi-Agent Financial Paper Trading Research Platform (Project AegisTrader)

Session Arc

The session moved through four distinct phases:

1. **Divergence**: Free Thinker proposed five creative directions, all centered on elevating the "trading soul" concept from supporting feature to product identity.
2. **Sorting**: Grounder evaluated all five — endorsed two strongly (T1, T4), refined one (T3), parked one (T2), cautioned one (T5).
3. **Deepening**: The endorsed threads gained specificity — onboarding flow, concrete counterfactual experiments, and a new coaching UX thread emerged.
4. **Convergence**: Arbiter confirmed three ideas as INTERESTING, forming a unified product narrative.

Inputs

- **Concept_1.md**: Detailed architectural blueprint — two-layer architecture, OpenClaw-inspired control plane, tech stack, trading soul structure, branch DAG, phased rollout.
- **PRD.md**: Full product requirements — 10 principles, agent authoring, backtesting, branching, approval-gated writes, live paper trading, 17+ data model entities, 5 major workflows.
- **Competitive landscape research** (two reports): Confirmed zero market analogs for soul evolution and branch DAG. Validated NautilusTrader as architectural reference. Identified QuantConnect Mia V2 as closest but fundamentally different competitor. Market size: $24B (2025) -> $44.5B (2030).

Outputs

Three Confirmed Ideas

1. Soul as the Product + Soul-First Onboarding Reframe AegisTrader from "backtesting platform with AI agents" to "platform for growing trading personalities that learn from experience." Core formulation: "Soul powered by backtest." Includes redesigned onboarding: archetype picker -> formative experience -> soul diff -> fork prompt.

2. Counterfactual Soul Forking (Trauma Test MVP) Fork beliefs, not just trades. MVP feature: Trauma Test — same agent, different eras, compare divergent souls side by side. V2: Selective Amnesia — remove a period, see which scars are wisdom vs damage. Introduces side-by-side soul comparison as a signature UX.

3. User-as-Coach with Soul Studio Replace binary approve/reject with coaching paradigm. Three modes: Let it Learn (passive, MVP), Guided Reflection (active, v1.5), Soul Surgery (direct, MVP). Soul Studio UI: timeline + belief cards + evidence drawer. Source tagging in soul.json. "Am I a Good Coach?" retention hook.

Unified Product Narrative

Choose a personality -> Grow it through experimentation -> Coach it to improve.

This is AegisTrader.

Thread Disposition

Thread	Final Status
T1: Soul as the Product	CONFIRMED INTERESTING
T2: Adversarial Soul Dynamics	Parked (phase 3/4)
T3: Soul Diff as First-Class UX	Absorbed into T1
T4: Counterfactual Soul Forking	CONFIRMED INTERESTING
T5: Soul-to-Deterministic Graduation	Cautioned; partially absorbed by T6
T6: Coaching UX / Soul Studio	CONFIRMED INTERESTING

Key Tensions and How They Resolved

1. **Soul as byproduct vs soul as product**: Resolved. "Soul powered by backtest" — soul is the UX surface, engine is the credibility layer.

2. **Deterministic boundary vs belief graduation**: Held. The coaching UX (T6) absorbs the useful parts of T5 without blurring the boundary — the user (not the system) decides what becomes a hard rule.
3. **Narrative vs evidence in soul diffs**: Productive tension. Direction: "narrative backed by evidence" — compelling but trustworthy.
4. **Game language vs domain language**: Resolved toward domain language. Archetypes named as trading philosophies, not character classes.
5. **Approver vs coach**: Resolved toward coaching. The user-soul relationship is developmental, not bureaucratic.

What Did Not Survive

- **Adversarial Soul Dynamics** (T2): Interesting but premature. Multi-agent game theory on top of PnL-only visibility adds complexity without helping define the core product.
- **Cross-Training Counterfactual** (T4c): Unclear user value. Academic curiosity, not a real user question.
- **Broad belief graduation** (T5): The broader vision of soul-as-hypothesis-generator feeding the deterministic engine was rejected. The narrow version (user approves a belief becoming a hard rule) is already in the PRD via approval-gated writes.

Impact on PRD

Sections to Rewrite

- Executive Summary: lead with soul evolution
- Vision: "grow AI trading personalities"
- User Stories: soul-curation as primary
- Onboarding: archetype picker flow
- UI: Soul diff as hero, Soul Studio as primary surface
- Approval-Gated Writes: expand to coaching modes

Sections to Preserve

- Deterministic engine architecture
- Branch DAG design (reframed, same technical model)
- Data sources and model stack
- File classification policy
- Non-functional requirements

New Elements

- Archetype templates (strategy.md + seed soul.md per archetype)
- Soul fork entity in data model
- Source tagging in soul.json schema
- CoachingInput, BeliefProvenance entities
- Coaching effectiveness metrics
- Side-by-side soul comparison UX
- v1.5 milestone for Guided Reflection

Artifacts Produced

- Ideation graph: `session/ideation-graph.md`
- Snapshots: `session/snapshots/SNAPSHOT_01.md` through `SNAPSHOT_05.md`
- Idea reports: `session/idea-reports/IDEA_soul-as-product.md`, `IDEA_counterfactual-forking.md`, `IDEA_coaching-ux.md`, `IDEA_user-as-coach-soul-studio.md`
- Briefs: `session/briefs/BRIEF_soul-as-product.md`, `BRIEF_counterfactual-forking.md`, `BRIEF_coaching-ux.md`
- Research: `session/research/COMPETITIVE_LANDSCAPE.md`, `RESEARCH_competitive-landscape.md`
- Vision document: `session/VISION_aegistrader.md`